



Evaluating Machine Learning Prediction Performance Across New York City Taxi Zones to Identify Underserved Communities

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ABSTRACT

There has been extensive research on taxi demand prediction in New York City based on machine learning; existing models only report overall accuracy scores that hide how poorly they actually perform in specific areas of the city. This project falls under the area of urban mobility and data analytics that solves three main issues: a lack of evaluation of the zone models in all 263 taxi zones in NYC, the inability of current prediction models to take into account contextual variables like the accessibility of subways and the car ownership rate, and the lack of an accessible planning tool to help transport authorities identify underserved areas. To overcome these issues, this project trains two ensemble tree-based machine learning models, XGBoost and Random Forest, on a set of NYC yellow taxi trip data between January and March 2025 and uses a zone-stratified evaluation framework to determine the individual accuracy scores of each of the 263 taxi zones. The results of two sets of features are compared: a baseline set based on the features of time and weather, and an enriched set with additional inputs of the proximity to subway stations and the car ownership rate. The results are displayed in a Python-created interactive dashboard using Plotly. It is hoped that the enriched model will show better performance in low-demand outer borough zones and that the dashboard will effectively reflect low-income communities with poor model performance and transit access. This project helps directly achieve SDG 11: Sustainable Cities and Communities, Target 11.2, which requires that everyone should have access to and affordable transport with special consideration to vulnerable groups. This project can assist the transport planners in making better and fairer decisions on the allocation of taxi resources in all five boroughs of New York City by generating zone-level evidence.

Keywords: taxi demand forecasting, machine learning, XGBoost, Random Forest, transport equity, SDG 11.

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CHAPTER 1: INTRODUCTION

1.1. Introduction

With an estimated 8.48 million residents as of July 2024, New York City (NYC) is one of the most populous metropolitan areas in the United States (New York City Department of City Planning, 2025). Manhattan, Brooklyn, Queens, the Bronx, and Staten Island are the five boroughs that make up the city, which is roughly 302.6 square miles in size (U.S. Census Bureau, 2024). One of the largest urban transit networks in the world facilitates the mass mobility of the population, with Manhattan, Brooklyn, Queens, and the Bronx as the four boroughs served by the NYC underground system (subway system), which is administered by the Metropolitan Transportation Authority (MTA, 2024). Staten Island is dependent on the Staten Island Railway, an independent commuter rail line run by the MTA, as it lacks a subway connection (MTA, 2024). In addition to rail transportation, the city is home to over 105,487 for-hire vehicles that are licensed by the NYC Taxi and Limousine Commission (TLC) to transport customers for a fee (NYC Taxi and Limousine Commission, 2025). These include ride-hailing services, which Correa and Moyano (2023) defined as app-based transportation services in which customers use a smartphone application to request trips and are instantly paired with a local driver. They also include green taxis and yellow taxis. These cars are vital in helping locals to reach their jobs, medical care, and other necessities.

Despite this, not every borough is equally served by the subway system. The MTA (2024) estimates that Manhattan has 121 subway station complexes covering about 23 square miles, resulting in a station density of about 5.3 complexes per square mile. In contrast, Queens has a density of only 0.7 complexes per square mile, with 79 complexes dispersed over roughly 109 square miles, whereas Staten Island has no subway station. Residents in the outer boroughs must travel much farther to reach the nearest subway station due to this disparity in station density. In the 45 most populous American metropolitan areas, Maharjan et al. (2024) looked at what they called the Modal Access Gap between transit and automobile, which is a score that captures how much worse public transit performs compared to automobiles, where the lower the score is, the better it is in helping residents reach their destination. They discovered that the gap ranges from 0.22 to 0.98, meaning that public

transit consistently fails behind the automobile in terms of access to employment opportunities

Additionally, their research revealed that communities with greater percentages of households without a car (a situation known as “careless households”) tend to be concentrated in the suburbs and exurbs of cities with the least amount of public transport (Maharjan et al., 2024). The issue is not limited to rail access. According to the Office of the New York City Comptroller (2025), citizens have fewer reliable options as bus services are less reliable. Evelly (2024) pointed out that a majority of New Yorkers are having trouble paying for public transport, which creates a financial barrier on top of the current spatial one. Overall, all of these factors demonstrate that people living in outlying zones continue to experience a mobility gap and that transportation access in NYC is still remarkably unequal.

In an effort to narrow those mobility gaps, several researchers have used machine learning techniques to forecast taxi demand and determine the distribution of taxi and ride-hailing pickups throughout the city. These demand forecasts will help in accurately predicting where and when demand for rides will be highest, and help transport planners to gain insights early on where demand will be more and provide more ride-hailing services there. Correa and Moyano (2023) defined machine learning as the use of algorithms that identify patterns from extensive spatiotemporal (**Spacio** means related to **space/location**, which says “where something is,” and temporal is related to time, which means “when something will happen”) data. These machine learning models include ensemble tree-based models and deep learning architectures. A specific deep learning model, such as the recurrent neural network called Long Short-Term Memory (LSTM) is made to recognize patterns in sequential data, for example, hourly taxi trip counts that are tracked over time. According to Mienye et al. (2024), an LSTM model works using three internal gates: the forget gate, the input gate, and the output gate. These gates help regulate which information the model needs to keep and which to discard at each time step. The forget gate determines which data from the preceding stage is no longer pertinent and eliminates it. Which recent data from the current step should be stored is determined by the input gate. What the model carries on to the next stage is then decided by the output gate. The model can identify patterns spanning lengthy sequences without losing crucial prior information due to this selective memory (Mienye et al., 2024).

Extreme Gradient Boosting, or XGBoost, is a realistic implementation of the gradient tree boosting model. It is formalized to control overfitting and produces trees consecutively, with each new tree rectifying the mistakes of the preceding ensemble (Correa & Moyano, 2023) to generate a single, more reliable forecast. Random forest is another ensemble of trained decision trees that mixes the output of several independently built trees (Ignatenko et al., 2024). To assess how predictable taxi demand is at each building block, Zhao et al. (2021) analyzed five prediction methods for taxi demand in Manhattan, including LSTM and Neural Network models. They utilized a metric known as temporal-correlated entropy. By taking into account both the frequency of each demand value and the order in which values develop across time, temporal-correlated entropy measures how regular a time series is; a lower entropy denotes a more predictable sequence.

Using a different set of algorithms, Correa and Moyano (2023) applied XGBoost and Random Forest to large-scale trip data from NYC and showed that ensemble tree-based models can perform well on tabular trip records, particularly when external features like location and weather are included. By using XGBoost and Random Forest on more than 90 million NYC trip records, Correa and Moyano (2023) showed the potential of these models. An RMSE of 38.51 and an aggregate R^2 of 0.97 were recorded by the XGBoost model and Random Forest produced an RMSE of 40.46 and an R^2 of 0.95, where the coefficient of determination, known as the R^2 , indicates how closely the prediction value of the model matches the actual values on a scale of 0 to 1. On the other hand, the RMSE (Root mean square error) is the average size of the prediction errors in the same unit as the target variable, where lower values denote greater accuracy (Chicco et al., 2021).

These numbers, however, mask a crucial flaw. Manhattan accounted for almost 86% of all taxi pickups among the 90 million trips in the dataset (Correa & Moyano, 2023). The model learns to optimize its predictions for the demand patterns of a particular region when the great bulk of training specimens are taken from that region. In the learning process, zones that contribute very few visits to the training set effectively become noise, meaning the model has little data to learn from when forecasting demand in those areas. In the field of spatial machine learning, Meyer and Pebesma (2022) illustrated this precise issue by demonstrating that a model with a strong overall metric may still consistently fail in particular regions that are under-represented in the training data, and that a single global accuracy score does not represent how a model performs at different locations. Because of this, the high aggregate R^2

score of Correa and Moyano (2023) may indicate good performance in Manhattan while concealing subpar accuracy in the surrounding boroughs.

This issue is shown from a different perspective in a related study. Manhattan was broken down into building blocks by Zhao et al. (2021), who then determined the most accuracy that any prediction algorithm could attain for each block. Markov, Lempel-Ziv-Welch, ARIMA, Neural Network, and LSTM were the five algorithms that were compared. The average maximum predictability of taxi demand across Manhattan building blocks was 83%, according to the study. With just 0.02% of the computation time required by LSTM, the Markov predictor attained 86% accuracy in regions with high maximum predictability (Zhao et al., 2021). Nevertheless, those findings were limited to Manhattan, the outputs reflected a theoretical upper limit rather than an actual demand estimate, and the geographical unit was the building block rather than the taxi zone. Neither the Zhao et al. (2021) study nor the Correa and Moyano (2023) research generated a per-zone prediction error or established any link between transit accessibility and prediction performance.

From this, we can address three gaps in the existing literature. First, none of the taxi zones in New York City have a machine learning model performance assessed at the zone level. The aggregate R^2 of 0.97 reported by Correa and Moyano (2023) does not demonstrate the true performance of the model in the outer boroughs in the absence of zone-level metrics. Second, no contextual features have been included in the prediction models of previous research. Contextual features are input variables that characterize the local conditions of a geographic zone. Examples include car ownership rate, which is the percentage of households in a zone that do not own a vehicle as reported in the American Community Survey (U.S. Census Bureau, 2024), or subway station proximity, which is the straight-line distance from the geographic center of a taxi zone to the nearest MTA subway station (MTA, 2024). Careless households are disproportionately concentrated in locations with poor transit coverage, as demonstrated by Maharjan et al. (2024), confirming that these factors capture significant disparities between zones. Third, no research to date has generated a useful tool that links transit accessibility evidence with zone-level prediction performance. By creating a repeatable framework that policymakers can utilize to identify socio-spatial disparities and carry out focused actions in support of SDG 11.

To solve this problem, this study will use XGBoost and Random Forest in a zone-stratified evaluation framework across all 263 NYC taxi zones. Instead of reporting a single aggregate score, a zone-stratified assessment methodology calculates model performance metrics independently for each geographic zone. The idea that predictions should be restricted to areas where the training data offers sufficient coverage and that spatial machine learning models should be assessed within their area of applicability was introduced by Meyer and Pebesma (2022). The framework of this study examines how well both models perform in high-demand and low-demand zones independently. The zones where the model performs poorly, and transit access is low, are then identified by combining the zone-level prediction error with transit accessibility metrics. The findings are displayed via an interactive dashboard that gives transport planners zone-level data to determine which zones need the deployment of taxi resources to be prioritized.

The project specifically aims to achieve **Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities**, specifically targeting SDG 11.2, which requires countries to give everyone access to safe, cheap, accessible, and sustainable transport networks, with a focus on the needs of those who are vulnerable (United Nations, 2015). The research provides transportation planners with the insights they need to prioritize deployment in marginalized communities by identifying NYC taxi zones with the least optimal model performance and lowest transit accessibility.

1.2. Problem Background

1.2.1 Problem Statement 1: Lack of zone-level machine learning model evaluation across all NYC taxi zones

Although New York City taxi demand prediction models have routinely returned high overall performance scores, none have examined how those scores hold up at the individual zone level across all 263 taxi zones in the city. In the field of spatial machine learning, Meyer and Pebesma (2022) showed that the performance of a model at many places is not represented by a single global accuracy score. In certain areas that are under-represented in the training data, a model may consistently fail while reporting a great overall statistic (Meyer and Pebesma, 2022). In the dataset utilized by Correa and Moyano (2023), Manhattan accounted for almost 86% of all pickups, making this issue particularly pertinent to taxi demand statistics. The model learns to optimize its predictions for a certain region when the great bulk of the training data is taken from that region. During the learning process, zones that contribute extremely few visits to the training set receive very little attention. As a result, the model lacks the information necessary to generate correct predictions for those locations.

After applying XGBoost and Random Forest to over 90 million trip records in NYC, Correa and Moyano (2023) reported aggregate testing scores of $R^2 = 0.97$ and $RMSE = 38.51$ for XGBoost and $R^2 = 0.95$ and $RMSE = 40.46$ for Random Forest. Neither score was broken down by borough or zone. Additionally, the study used 263 zone dummy variables as model features, which suggests that instead of learning generalizable correlations between the input characteristics and demand, the model could successfully recall zone-specific demand patterns (Correa & Moyano, 2023). A reader of their paper can't ascertain whether the high aggregate score is exclusive to Manhattan or if the model performs similarly throughout the city.

By calculating theoretical predictability limitations for every Manhattan building block, Zhao et al. (2021) investigated prediction performance at a finer spatial scale. Nevertheless, their analysis was limited to Manhattan. The building block was utilized as the spatial unit instead of the taxi zone, and the results were a theoretical upper bound on prediction accuracy rather than a real demand estimate. Per-zone prediction performance across all 263 NYC taxi zones has not been reported in any study in the literature to date.

Transport planners are unable to determine which communities are being served by an erroneous model without these metrics. SDG 11 target 11.2 (United Nations, 2015) is directly hampered by this since people of outer boroughs who rely mostly on taxis cannot be identified as underserved if their zones are never evaluated independently.

1.2.2 Problem Statement 2: Absence of contextual features in existing taxi demand prediction models for outer borough zones

The lack of contextual characteristics in the input data prevents a model from explaining why demand varies between zones, even if it receives a good overall score. The demand for taxis is influenced not just by the time of day but also by the public transportation accessibility and car ownership rates of a zone. In their analysis of NYC for-hire vehicle records from 2019 to 2023, Lie and Chen (2026) found a notable center-periphery gap, with wealthy central neighbourhoods remaining steady while outer borough neighbourhoods frequently relied on for-hire vehicles for everyday transportation. Households without a car are disproportionately concentrated in suburban and exurban areas with the lowest transit coverage, according to the analysis of Maharjan et al. (2024) for transit and automobile access gaps across the 45 most populous American metropolitan areas. These results verify that conditions that are fundamentally different from those in Manhattan influence taxi demand in outlying borough zones.

The two main studies on taxi demand in NYC, however, relied nearly exclusively on atmospheric and temporal characteristics in spite of this data. Zone code, hour of day, day of week, temperature, and precipitation were the only five features Correa and Moyano (2023) included in their model. Both demographic characteristics, like car ownership rates and transportation accessibility factors, were absent. In their section on future work, the authors explicitly acknowledged this limitation, pointing out that it was never investigated whether outer borough taxi pickups cluster close to subway stations (Correa & Moyano, 2023). Zhao et al. (2021) employed a slightly more comprehensive collection of weather features, which included wind speed but excluded demographic data and transit proximity. The factors that influence demand in outlying borough zones were not included in either study.

This restriction is particularly significant in low-demand areas where temporal features are insufficient. The program can learn demand patterns from time-based variables alone because Manhattan zones observe thousands of past trips every hour. The algorithm has relatively little data to learn from and no contextual characteristics to make up for it in outlying borough zones with sparse and erratic travel records. Accuracy thus deteriorates the most in such zones. This relates to SDG 11 target 11.2 (United Nations, 2015), which mandates that planners provide inclusive transport coverage based on evidence derived from data.

1.2.3 Problem Statement 3: Lack of an actionable tool for transport planners to identify and address underserved zones

No previous study has integrated contextual data on transit access and zone-level evidence of model failure into a single tool that transport planners can use. Planners require both zone-level information on where current models fall short and contextual evidence of why particular zones are more difficult to service to distribute taxi resources equally. Without both, possibilities on where to add more services are based on insufficient data. By creating a repeatable framework that policymakers can utilize to identify socio-spatial imbalances and carry out specific actions in support of SDG 11, Liu and Chen (2026) illustrated the importance of integrating mobility data with neighbourhood context. There has not been a comparable technique for forecasting taxi demand.

Correa and Moyano (2023) acknowledged that their models may provide city planners and taxi operators with information about where to put taxis during the day. Their paper finished with aggregate scores. There was no visual output that a planner could utilize to compare performance across the 263 zones, no transit accessibility context, and no zone-level breakdown. Zhao et al. (2021) linked land use in Manhattan to predictability, but they did not provide an analysis that planners could employ for the entire city. Although the evidence required to solve transport inequity is present in many databases and research papers, it has never been compiled in a way that would allow city authorities to take action.

1.3. Project Aim

To evaluate machine learning model prediction performance at the individual taxi zone levels across all five boroughs of New York City using XGBoost and Random Forest with contextual transit and demographic data, to assist transport planners in determining which communities require increased taxi services.

1.4. Objectives

1. To evaluate the zone-level prediction performance of XGBoost and Random Forest models trained on temporal and weather features across all 263 New York City taxi zones, computing individual Root Mean Squared Error (RMSE) and R-squared (R^2) scores for each zone as a baseline for comparison.
2. To assess whether incorporating contextual features, such as subway station proximity and car ownership rate, improves model prediction accuracy in low-demand outer borough zones compared to the baseline model.
3. To produce an interactive dashboard that integrates zone-level prediction performance metrics from the enriched model alongside transit accessibility indicators, to help transport planners identify which communities across all five boroughs require prioritized taxi service deployment.

1.5. Scope

1.5.1. What will be covered for this project

1.5.1.1 Study Area and Spatial Unit

- The geographical areas that will be covered for this study are all five boroughs of New York City (NYC), which include:
 - Manhattan
 - Brooklyn
 - Queens
 - the Bronx
 - Staten Island
- The spatial unit of analysis is the 263 official NYC taxi zones as defined by the NYC Taxi and Limousine Commission (TLC) shapefile.

1.5.1.2 Data Collection

Five open-source government datasets will be collected for this study:

- **NYC TLC Yellow Taxi Trip Records** (January to March 2025): trip-level pickup records for every yellow taxi journey made in each of the five boroughs from January to March 2025.
- **NYC TLC Taxi Zone Shapefile**: Record of geographic polygon boundaries for all 263 official NYC taxi zones
- **MTA Subway Station Data**: Record of geographic coordinates of all MTA subway stations in New York City
- **NOAA Weather Data**: hourly temperature and precipitation records from January to March 2025 from the Central Park station (WBAN: 94728).
- **US Census ACS Table B08141**: American Community Survey 5-Year 2024 projections for zero-vehicle households proportions at the census tract level.

1.5.1.3 Data Preprocessing

To create a single zone-hour dataset fit for modelling, each of the five datasets will be cleaned, integrated, and aggregated. Null value handling, invalid trip elimination, temporal feature extraction, hourly aggregation per zone, spatial joins between datasets at the zone level, and final dataset validation are all included in the preprocessing.

1.5.1.4 Feature Engineering

- **Baseline Feature Set** (temporal and weather features only)
 - Hour of day (0 to 23)
 - Day of week (0 to 6)
 - Month (1 to 3)
 - Temperature in Fahrenheit
 - Precipitation in inches
- **Enriched Feature Set** (all five baseline features plus the following two additions)
 - **Subway station proximity**: straight-line distance in kilometres from zone centroid to nearest MTA station
 - **Car ownership rate**: calculated as the straight-line distance in kilometers from the zone centroid to the closest MTA station

1.5.1.5 Model Training and Evaluation

The baseline feature dataset and the enriched feature dataset will be used to train two ensemble tree-based models twice. Random Forest and XGBoost are the two models. In accordance with the methodology of Correa and Moyano (2023), the data is divided chronologically, with 80% being utilized for training and 20% for testing. At this point, both models employ the default hyperparameters. After that, the models are assessed independently for zones with high and low demand.

1.5.1.6 Dashboard Development

- An interactive dashboard will be created as the deliverable.

- The dashboard will display the following information for each of the 263 NYC taxi zones:
 - Zone-level R^2 and RMSE from the baseline model
 - Zone-level R^2 and RMSE from the enriched model
 - Subway station proximity value for each zone
 - Car ownership rate for the zone
- Zones where both models show poor R^2 and high RMSE alongside low transit accessibility will be visually flagged as underserved zones
- The dashboard will allow transport planners to compare model performance across zones and identify communities that require priority taxi resource deployment
- The dashboard is a standalone data visualization output built in Python using Plotly or an equivalent library and is not a deployed web application

1.5.2 Constraints

1.5.2.1 Data Constraints

- Trip data is limited to three months (January to March 2025) due to computational feasibility within a single-semester dissertation timeline
- Weather data comes from a single Central Park station and does not capture localized weather variation across the five boroughs
- Census ACS data reflects the 2019 to 2024 reference period and may not perfectly represent car ownership conditions in early 2025
- Census tract boundaries do not align perfectly with taxi zone boundaries, which may introduce minor imprecision in the area-weighted aggregation of car ownership rates

1.5.2.2 Modelling Constraints

- Default hyperparameters are used for all four models; no hyperparameter optimization is applied at this stage
- The model predicts hourly pickup counts only and does not predict drop-off demand, waiting times, or real-time fluctuations
- Very low-demand zones with sparse test set records may produce statistically unreliable R^2 scores and will be flagged in the results

- The model is trained on historical data only and does not support real-time prediction
- The project is developed by a single researcher within a single semester, thus limiting the range of models that can be implemented and tested. Only two models are evaluated, and hyperparameter tuning is constrained to default settings at this stage.

1.5.2.3 Spatial Constraints

- Subway proximity is measured as straight-line distance and does not account for walking routes, road networks, or physical barriers such as rivers
- Zone centroids are used as representative points for each zone and may not accurately represent average travel distances in large or irregularly shaped outer borough zones

1.5.2.4 Technology Constraints

- All processing and modelling will be performed on a local machine without a cloud computing infrastructure, which constrains the data volume that can be processed
- All implementations will be in Python only using pandas, scikit-learn, GeoPandas, and Plotly

1.5.2.5 System Constraints

- The dashboard is a standalone visualization deliverable and does not include real-time data feeds, user authentication, or deployment infrastructure
- The system does not provide prescriptive recommendations; it presents zone-level evidence for transport planners to act on

1.5.2.6 Time Constraints

- The full project must be completed within Semester 2 of the 2026 academic year
- All major decisions are subject to supervisor confirmation within a minimum of three required meetings per semester

1.5.3. What will not be covered in this project

- **Green taxi data** is excluded because green taxis were designed to serve the outer boroughs, and including them would reduce the Manhattan-skew imbalance that this study is specifically designed to investigate.
- **Ride-hailing data**, including Uber and HVFHV records, is excluded to maintain a focused comparison with the yellow taxi baseline established by Correa and Moyano (2023)
- **Zone ID (PULocationID)** is not included as a feature in the model training since it would cause the model to memorize zone-specific demand patterns rather than learn from meaningful temporal and contextual features, even though this feature is used as a join key during preprocessing and as a grouping variable during zone-level evaluation. It is deliberately removed since the Correa and Moyano (2023) paper included zone ID as dummy variables in their model and reported an aggregate R^2 of 0.97 with no zone-level breakdown. This makes it unclear whether their score reflects genuine prediction accuracy or simply the model having memorized Manhattan zone labels.
- **Deep learning models** such as LSTM and CNN are out of scope; only ensemble tree-based models are evaluated since this study focuses on ensemble tree-based models to maintain direct comparability with Correa and Moyano (2023) paper. Deep learning models are a direction for future work
- **Real-time prediction** is not included since the study uses historical records only to understand historical patterns. A real-time deployment would require a live data pipeline, which is outside the scope of a single-semester dissertation.
- **Taxi drop-off demand** is not predicted because the target variable is pickup count only, which is focused on Correa and Moyano (2023) for direct methodological comparison.
- **Population density and median household income** are not included as model features to avoid redundancy because the car ownership rate already directly captures vehicle access at the zone level.
- **Wind speed and other weather variables** beyond temperature and precipitation are excluded because only the two weather features were used by Correa and Moyano (2023) to maintain a controlled baseline comparison

- **Bayesian optimization** is not applied at this stage because default hyperparameters are used to ensure that performance differences between the baseline and enriched models are attributable solely to the feature sets. On the other hand, a simple grid search may be explored for the enriched model only in Semester 2 if needed
- **Route optimization and fleet dispatching** are not addressed because this project produces demand forecasts and zone-level insights only and does not prescribe operational decisions to taxi operators.
- **Other cities or regions** outside New York City are not included because both anchor papers by Correa and Moyano (2023) and Zhao et al. (2021) have conducted their research exclusively within New York City, and this study is designed to directly build on and critique their work within the same geographic context.

1.5.4. Target Users

- **Urban transport planners and city authorities** in New York City who require zone-level demand forecasts and transit accessibility evidence to make data-driven decisions about where to prioritize taxi resource deployment across all 263 NYC taxi zones
- **Data science researchers** in the urban mobility domain who may reuse the enriched dataset or zone-stratified evaluation framework to extend this work to other model types, time periods, or cities

1.6. Potential Benefits

1.6.1 Tangible benefits

- **Zone Level prediction metrics:** Transport planners receive R^2 and RMSE scores computed individually for each of the 263 NYC taxi zones. This will replace the misleading single aggregate score reported by Correa and Moyano (2023).
- **Identification of underserved zones:** This will allow transit planners to have a full list of zones where model performance is poor and transit accessibility is low, allowing city planners to identify which specific communities are most in need of additional taxi and ride-hailing resources that directly support SDG 11 Target 11.2.

- **Reusable enriched dataset:** The project will produce a merged dataset combining NYC TLC yellow taxi trip records with MTA transit, NOAA weather, and Census car ownership data at the zone level for all 263 zones. This dataset can then be used by other researchers for their future urban mobility research work without having to repeat the data collection and preprocessing pipeline.
- **Interactive Dashboard:** A visual tool will be made to visualize zone-level model performance and transit accessibility indicators across all 263 zones in one place, eliminating the need for cross-referencing multiple separate datasets and reports when making resource deployment decisions.

1.6.2 Intangible benefits

- **Improved transport equity:** Evidence-based identification of underserved outer borough communities that supports more equitable taxi resource deployment across New York City by identifying which outer borough zones have both poor model accuracy and low transit accessibility, which supports the needs of residents in communities that are currently underserved by both public transit and on-demand transport services.
- **Advancement of evaluation practices:** The zone-stratified framework challenges common practices of reporting only aggregate model performance scores in urban mobility research. If adopted by future researchers, this approach would produce more honest and actionable accuracy assessments across diverse spatial contexts.
- **Contribution to SDG 11 Target 11.2:** Provides city authorities with the zone-level evidence needed to prioritize transport deployment in communities with the weakest transit access.
- **Foundation for future research:** The methodology and dataset provide a replicable starting point for researchers to extend this work to other cities, time periods, or model types, such as ensemble tree-based models or deep learning approaches at the zone level.

1.7. Overview of the IR

1.7.1 Chapter 1: Introduction

This chapter presents the machine learning field of taxi demand prediction in New York City. It provides the problem background by stating three problem statements, which include a lack of zone-level model analysis, a lack of contextual characteristics in the current models, and the lack of an actionable planning instrument. The chapter outlines the project objective, three quantifiable goals, scope limits, target audience, possible benefits, and a general project outline.

1.7.2. Chapter 2: Literature Review

This chapter discusses the field research on the areas of taxi demand prediction and urban mobility. It also contains a critical analysis of related works and the in-depth analysis of the two anchor papers by Correa and Moyano (2023) and Zhao et al. (2021) and other similar works. The chapter ends with a discussion on technical research, including the tools, libraries and software environments chosen to use in the project.

1.7.3. Chapter 3: Methodology

This chapter describes the data analytics approachology to be used in this project, such as a comparison between CRISP-DM, KDD and SEMMA, and the reason behind the choice of approach. It also records the five datasets that were collected, the entire data preprocessing pipeline used on each dataset and the initial data understanding that was conducted on the cleaned and integrated dataset.

1.7.4. Chapter 4: Conclusion

The last chapter is a discussion of the success of the investigation phase and how much the project objectives have been achieved. It links the results to SDG 11 Target 11.2 (United Nations, 2015) and the research gaps and areas of the design that will be filled in Semester 2.

1.8. Project Plan

PROJECT PLAN				
Task Name	Duration (Days)	Start Date	End Date	Status
Project Proposal Form (PPF)	16	26 Jan 2026	11-Feb-26	Completed
Chapter 1: Introduction	7	12 Feb 2026	18 Feb 2026	Completed
1.1 Introduction	1	12 Feb 2026	12 Feb 2026	Completed
1.2 Problem Background	1	13 Feb 2026	13 Feb 2026	Completed
1.3 Project Aim	1	14 Feb 2026	14 Feb 2026	Completed
1.4 Objectives	1	14 Feb 2026	14 Feb 2026	Completed
1.5 Scope	1	15 Feb 2026	15 Feb 2026	Completed
1.6 Potential Benefits	1	16 Feb 2026	16 Feb 2026	Completed
1.7 Overview of the IR	1	17 Feb 2026	17 Feb 2026	Completed
1.8 Project Plan	1	18 Feb 2026	18 Feb 2026	Completed
Chapter 2: Literature Review	14	19 Feb 2026	04 Mar 2026	Completed
2.1 Introduction	1	19 Feb 2026	19 Feb 2026	Completed
2.2 Domain Research	8	20 Feb 2026	27 Feb 2026	Completed
2.3 Similar Works	3	28 Feb 2026	02 Mar 2026	Completed
2.4 Technical Research	1	03 Mar 2026	03 Mar 2026	Completed
2.5 Summary	1	04 Mar 2026	04 Mar 2026	Completed
Chapter 3: Methodology	21	05 Mar 2026	25 Mar 2026	Completed
3.1 Introduction	1	05 Mar 2026	05 Mar 2026	Completed
3.2 Methodology	5	06 Mar 2026	10 Mar 2026	Completed
3.3 Data Collection	4	11 Mar 2026	14 Mar 2026	Completed
3.4 Data Pre-processing	10	15 Mar 2026	24 Mar 2026	Completed
3.5 Summary	1	25 Mar 2026	25 Mar 2026	Completed
Chapter 4: Conclusion	4	26 Mar 2026	29 Mar 2026	Completed
4.1 Achievement	1	26 Mar 2026	26 Mar 2026	Completed
4.2 Justification (SDG)	2	27 Mar 2026	28 Mar 2026	Completed
4.3 Research Gaps	1	29 Mar 2026	29 Mar 2026	Completed
References (APA Style)	2	30 Mar 2026	31 Mar 2026	Completed
Appendix A: PPF Screenshots	1	01 Apr 2026	01 Apr 2026	Completed
Appendix B: Ethics Form	1	01 Apr 2026	01 Apr 2026	Completed
Appendix C: Log Sheets	1	02 Apr 2026	02 Apr 2026	Completed
Appendix D: Gantt Chart	1	02 Apr 2026	02 Apr 2026	Completed
Proofreading & Formatting	5	06 Apr 2026	10 Apr 2026	Completed
Supervisor Review & Revisions	7	11 Apr 2026	17 Apr 2026	Completed
Final Checks & Binding	2	18 Apr 2026	19 Apr 2026	Completed
IR Submission	1	22 Apr 2026	22 Apr 2026	Completed

Figure 1: Project Plan

CHAPTER 2: LITERATURE REVIEW

2.1. Introduction

This chapter is a literature review on the current literature that is applicable in the development of a zone-stratified taxi demand prediction framework of the city of New York. The review is organised into four sections. Section 2.2 introduces the domain research which discusses four important points that are: the urban transport systems and taxi demand in NYC, machine learning applications in taxi demand forecasting, feature engineering and dataset design, and model testing in spatially structured data. Section 2.3 provides a comparison of seven close works that best relate to this project. Section 2.4 is a review of the technical tools and technologies used in the project. Section 2.5 presents an overview of the main findings of the literature review and also points out the gaps that this project aims to fill.

2.2. Domain Research

2.2.1 New York City urban transport systems and taxi demand.

2.2.1.1 The urban transport systems and the role taxis and for-hire vehicles play in them.

Taxis and for-hire vehicles are crucial links in an urban transport system. They offer door-to-door accessibility, which the fixed route public transit cannot offer. This is particularly crucial in localities where there are limited bus and rail services. Cats et al. (2022) examined the Uber trip data in six cities (United States and Europe) and discovered that 20-40 percent of all ride-hailing journeys did not have a feasible public transit alternative. A viable public transit option implies that there is a bus or train option they can use that can carry the passenger to their pickup point in a reasonable time (Cats et al., 2022). This result proves that vehicle-on-demand is not a mere replication of the existing transit services. They occupy real gaps in the transport system where there is no viable public transport.

Cats et al. (2022) came to the conclusion that the correlation between ride-hailing and public transit cannot be condensed to the question of either competition or support. They found out that the two occur simultaneously, but they are dependent upon areas. This trend was supported by Jin et al. (2019) in the case of New York City. They discovered that Uber is competing with public transit in the majority of the daytime hours in places that have good transit coverage. Nevertheless, Uber supplements the public transportation at midnight and in areas where the transportation service is inadequate (Jin et al., 2019). This implies that the demand of ride-hailing is influenced by various factors in varying locations of the city. Ride-hailing is convenient for passengers in Manhattan. In the outer boroughs, passengers prefer ride-hailing, as they have limited alternatives.

Flor et al. (2022) twisted this argument. Their analysis of ride-hailing services in Madrid revealed that ride-hailing services enhance the availability of transportation to customers who experience challenges getting to taxis or public transportation points. This covers residents in communities around the periphery of the city where there is low access to services. Nevertheless, they also revealed that ride-hailing did not add to but mostly replaced urban bus services (Flor et al., 2022). Combined, these studies confirm the existence of a two-sided role that taxis and for-hire cars play in urban transport. They act as a required mode of transport to the people living in regions with poor transportation. Meanwhile, they are competing with public transit in the regions where buses and trains are present.

2.2.1.2 Transport landscape in New York City and the outer borough service gap.

The transport system of New York City can be characterized by an apparent imbalance between Manhattan and the four outer boroughs. Manhattan, Brooklyn, and the Bronx are served by the subway system run by the Metropolitan Transportation Authority (MTA), which lacks the extension to Staten Island (MTA, 2024). The density of the subway stations per square mile in these four boroughs ranges far and wide. As identified in Chapter 1, Manhattan has around 5.3 subway station complexes per square mile, as compared to Queens with 0.7. There are no subways in Staten Island. This implies that people living in Queens and Staten Island will have to travel a long distance to access the closest subway station compared with people in Manhattan.

The same imbalance is evident in the taxi system in New York City. Taxi yellow cars can take on board passengers at any point in the city, but in street hails, meaning passengers who do not book a taxi with a phone application but hail it on the street (Correa and Moyano, 2023). According to Correa and Moyano (2023), yellow taxis tend to work in the most densely populated regions of the city, that is, the Central Business District of Manhattan and two large airports. Since the natural demand for hailing on the street is the busiest in areas with the greatest number of people, the concentration of yellow cabs in Manhattan is not by law but by choice. Consequently, Correa and Moyano (2023) discovered that yellow taxis represented 86% of all taxi pickups in their sample, with the overwhelming majority of them being located in Manhattan.

In order to fill the service gap, the NYC Taxi and Limousine Commission (TLC) came up with green taxis in 2013. Green taxis are another type of for-hire vehicles that are specifically created to serve the outer boroughs and northern Manhattan, where the supply of yellow taxis is often low (Correa & Moyano, 2023). Xie et al. (2021) affirmed that the green taxis were introduced to address the issue of a lack of taxi services beyond the downtown area. In spite of this intervention, the gap in services remains. With 2014 and 2015 trip records, Correa and Moyano (2022) compared the Uber growth with that of green taxis in the outer boroughs. They discovered that both services were increasing, although at very dissimilar rates. The increase in Uber travel in the outer boroughs was 550 percent over the time of the study, and the green taxi travel increased by only 29 percent (Correa and Moyano, 2022). This is because even as the green taxis continued to attract more riders, Uber was acquiring new riders almost 19 times more rapidly.

Barajas and Brown (2021) inquired about the ability of ride-hailing services to address gaps in what they described as transit deserts. A transit desert is a region where demand for transit services is high, yet the supply of transit services is low (Barajas & Brown, 2021). In their research, they examined whether ride-hailing penetrates the locations that are not well covered by public transit. Results in their study affirm that the outer borough service gap is a long-lasting issue that has not been completely addressed by green taxis and ride-hailing services. It is this gap that makes the demand for taxis in the outer boroughs important to predict for transport planning.

2.2.1.3 For-hire vehicle patterns of demand by borough.

Liu and Chen (2026) examined records of for-hire vehicles in NYC in 2019-2023 and discovered that more prosperous areas close to the city centre had no significant changes in travel demand throughout the years. Conversely, the more remote locations grew, and residents became more and more reliant on for-hire vehicles to reach their destinations. Liu and Chen (2026) referred to this trend as a centre-periphery divide, which implies that the disparity between transport choices in the city centre and the suburbs increased during the period of the research.

This issue was investigated by Maharjan et al. (2024) on a national level. They quantified the so-called Modal Access Gap (MAG) in the 45 largest American metropolitan areas. The Modal Access Gap is a comparison of the number of jobs accessible by someone using public transit to the number of jobs that the individual person can access using a car (Maharik et al., 2024). The outcome is an integer of 0 to 1. A score near 0 indicates that there is little difference between transit and driving in accessing jobs. Values that are nearly 1 indicate that driving allows access to a much larger number of jobs compared to transit. Maharjan et al. (2024) discovered that the MAG was between 0.22 and 0.98 in the 45 cities. None of the cities had a score near 0. This implies that driving in all the cities they investigated offered more access to employment compared to public transport. Their analysis also concluded that households without a car are also concentrated in the suburbs with the least transit coverage. This is a scenario where the population most in need of transit is in the areas where transit is worst.

Correa and Moyano (2022) discovered that the growth trend in the outer boroughs was not homogeneous. The Uber trips increased by 528 percent in Brooklyn, 1,216 percent in the Bronx, 530 percent in Queens, and 1,156 percent in Staten Island throughout the study period (Correa & Moyano, 2022). These figures indicate that the outer boroughs cannot be considered one group with the same travel behaviour. Different boroughs have their demand patterns based on their transit infrastructure and availability of for-hire vehicles. This is exactly the reason why there can be no single aggregate model performance score that can reflect the level of service of a prediction

model to each borough. A model with a high total score can be low in certain outer borough areas where demand does not match Manhattan.

2.2.2 Taxi demand prediction using machine learning.

2.2.2.1 General overview of machine learning in transportation demand forecasting.

Machine learning is a category of computer software that identifies trends in extensive data and applies the trends to draw predictions about new data that it has never encountered (Sarker, 2021). When applied to taxi demand prediction, the program can learn based on the records of past trips and rely on the trends it identifies to predict the number of taxi pickups in the area at a certain time. Correa and Moyano (2023) used XGBoost and Random Forest as their models to apply machine learning algorithms to infer the demand distribution of taxi and Uber pickups across New York City.

The methods of machine learning that are used in taxi demand prediction can be classified into three. The former is the statistical models, which are prediction techniques that involve fixed mathematical equations to describe the relationship between the input variables and the output. They can be ARIMA, which forecasts future values using previous values in a time series, and linear regression, which uses a straight line through the data to predict the output (Peng et al., 2023). The second type is traditional machine learning models, a type of algorithm that is trained on the relationship between the inputs and outputs using the data instead of a pre-defined equation, which is to be entered into the algorithm by the user (Sarker, 2021). Examples are Random Forest, XGBoost and Support Vector Machine. The third category is deep learning models, a more complicated form of machine learning that involves several layers of processing units arranged one over the other to identify patterns in data (Mienye et al., 2024). They can be Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN). Peng et al. (2023) performed a systematic review whose targeted deep learning approaches are taxi demand prediction. They have looked through the works published 2017-2023 and talked about the merits and shortcomings of various deep learning structures to process the space and time dimensions of taxi trip data.

Alam et al. (2025) applied four machine learning models to a dataset at the New York City Taxi and Limousine Commission. They contrasted generalized linear regression (GLR), a form of regression model that builds on ordinary linear regression to permit the output variable to exhibit various types of statistical behaviour, not simply a straight line (Alam et al., 2025), with multinomial logistic regression, Random Forest, and Decision Tree. They found that GLR had the best performance with an R-squared score of about 90. Random Forest and Decision Tree were worse than GLR, which Alam et al. (2025) explained by overfitting due to the presence of extremely imbalanced data of demand. Overfitting is a scenario whereby a model is too fit to the training data and learns patterns of the training data instead of patterns relevant to the new data (Alam et al., 2025). Here, the demand in most zones was very low and in a few zones was very high, and this made the tree-based models overfit in the prevalent low-demand patterns.

This finding is contrary to Correa and Moyano (2023), who found that the R-squared of Random Forest was 0.95 on an equivalent dataset of NYC taxis. The two studies, however, employed different input features. Correa and Moyano (2023) used 263 zone dummy variables as input features, where the model directly learned the level of demand of each zone using the zone label. Alam et al. (2025) have not used the zone identity features in this manner. This disparity in feature design could be one of the reasons why the results of the two studies with the use of Random Forest were different. The impact of the zone identity features on the model performances is further discussed in Section 2.2.3.4.

2.2.2.2 Deep learning models and limitations on tabular trip data.

Taxi demand prediction has been a wide-range application of deep learning models due to their nature of modelling complex trends in both time-sequential and spatial data. Long Short-Term Memory (LSTM): This is a form of recurrent neural network that is used to learn patterns in a sequence of data, e.g. the number of taxi trips per hour over time. Mienye et al. (2024) described that an LSTM has three internal gates, namely, the forget gate, the input gate, and the output gate. The forget gate determines which information from the last step is not important anymore and eliminates it. The input gate determines the new information of the current step that should be stored. The output gate then dictates what the model forwards to the next step. Such selective memory enables the model to learn the patterns that cover long sequences without forgetting valuable information in the past (Mienye et al., 2024). Convolutional Neural Networks (CNN) are an alternative model of a deep learning model that was initially intended to perform image recognition. In taxi demand prediction, CNNs consider a city grid as an image with each cell acting as a zone, and the cell value is a number of pickups in this zone at a certain time window (Peng et al., 2023).

Zhao et al. (2021) have compared five prediction algorithms in Manhattan (Neural Network and LSTM) on taxi demand. Their results revealed an average maximum predictability of taxi demand of 83% in Manhattan building blocks. But in high predictability, they also discovered that a Markov predictor was 86% accurate. Markov predictor is a probabilistic algorithm which predicts the next value in a sequence by examining the last value in the sequence and determining the most probable value based on the frequency of each transition occurrence in the past (Zhao et al., 2021). It is not a machine learning nor a deep learning model; it is a statistical method. Although it was significantly simpler than LSTM, the Markov predictor only needed 0.02% of the computation time LSTM required, but it was more accurate in high-predictability regions (Zhao et al., 2021). This finding indicates that deep learning is not necessarily the most effective in terms of accuracy and time spent training the model. The authors Zhang et al. (2021) suggested a Multi-Level Recurrent Neural Network (MLRNN) that introduced a solution of zone heterogeneity by clustering taxi zones in NYC according to demand correlation. They modelled them with independent recurrent neural networks per cluster to model the distinct demand patterns in each cluster. Although this method

enhanced the accuracy of predictions, it made the model more complex and necessitated a multi-stage training (Zhang et al., 2021).

One of the major constraints of deep learning models can be well understood when the input data is not a sequence or an image, but in tabular form. Tabular data are data that are arranged in rows and columns like a spreadsheet. The rows are the records and the columns are the features (Grinsztajn et al., 2022). Here is one example: a taxi trip dataset is tabular as each row corresponds to a zone-hour combination and the columns include the features describing the hour of the day, temperature, and pickup count.

Grinsztajn et al. (2022) carried out a benchmark analysis of 45 datasets in various fields to learn why tree-based models always perform better than deep learning on this kind of data. Their research found three particular reasons. To begin with, tree-based models are resistant to uninformative features, i.e., input columns that, in fact, do not assist in predicting the output (Grinsztajn et al., 2022). Tree-based models saw a smaller performance decline when half of the least important were removed in each dataset, whereas deep learning models experienced a bigger performance drop. This is important to taxi demand prediction since not all of the features in a dataset will be effective in all zones.

Second, tree-based models maintain the orientation of the data, i.e. they base their decisions on considering one feature at a time (Grinsztajn et al., 2022). The values in each column of a tabular dataset are independent. This structure is observed to be respected by tree-based models since each split in the tree only takes into account a single feature. Third, tree-based models are able to discover non-regularities in the data, including drastic changes or abrupt shifts in the value of the output when a feature reaches a particular threshold (Grinsztajn et al., 2022). Deep learning models are more likely to learn smooth patterns and do not cope with the sharp edges that tree-based models are more adept at.

These findings were supported by Chou et al. (2023) in the case of taxi demand. They observed that big deep models involving transformers and BERT-based models are computationally intensive when using them in applications that predict commercial taxi demand. They found that mid-level models such as LSTM combined with ensemble

learning provided a better balance between accuracy and computational cost (Chou et al., 2023). Such results, along with the evidence presented by Grinsztajn et al. (2022), give three distinct reasons why ensemble tree-based models should be used in this project, as opposed to deep learning. The NYC TLC trip data is in the form of a table. It has properties whose meaning is independent that are naturally dealt with in tree-based models. And tree-based models are resistant to uninformative features, useful where some features might only be useful in specific regions.

2.2.2.3 Tree-based taxis demand prediction ensemble model.

Ensemble tree-based models are machine learning models that utilize several decision trees to generate a more accurate prediction (Ignatenko et al., 2024). A decision tree is a model that divides the data into small collections depending on a set of yes-or-no inquiries about the attributes of the input features. The data is divided into two branches, with each split, and the process is repeated until a final prediction is obtained at the end of the model (Ignatenko et al., 2024). Random Forest and XGBoost are two models that will be employed in this project.

Random Forest is a collection of trained decision trees, which are independent. All the trees in the ensemble are trained on a random sample of the training data, a method known as bootstrap sampling. Bootstrap sampling implies that each tree gets a random part of the training records, with some possibly being repeated several times and others being omitted altogether (Ignatenko et al., 2024). This implies that every tree trains on a variant of the data that is slightly different. Each tree also only looks at a random subset of the available features at each split point. The average of all the outputs of the individual trees forms the final prediction (Ignatenko et al., 2024). This randomization of training data and randomization of feature selection makes Random Forest resistant to overfitting, i.e. less likely to memorize the training data and more likely to be useful on new data that it has never encountered.

XGBoost or Extreme Gradient Boost operates in a different manner as compared to Random Forest. XGBoost does not train all trees in parallel, but instead it trains trees sequentially. The new trees are trained to correct the mistakes that the old trees have made. It implies that the model pays attention to the records which were the hardest to anticipate in the last round. According to Correa and Moyano (2023), XGBoost is a

convenient implementation of the gradient tree boosting model formalised to control overfitting. The model incorporates inbuilt checks that keep it from being excessively close to the training data, which in turn assists it to work well on unseen test data (Correa and Moyano, 2023).

The two models have been used in predicting taxi demand, with good results. Correa and Moyano (2023) trained XGBoost and Random Forest on a total of more than 90 million NYC trip records and found an overall R squared of 0.97 and 0.95, respectively. In its study, Gangrade et al. (2022) applied ensemble regression techniques such as the Random Forest to Chicago taxi data that incorporated the time, location, and social-demographic characteristics. They discovered that the use of multiple input features, which included temporal features, also led to better prediction results than temporal features alone (Gangrade et al., 2022). The independent comparison of taxi demand models identified that both Random Forest and XGBoost models performed better than the baseline models because of their capability to address the complicated patterns in the urban taxi demand data (ISI, 2024). These results justify the choice of XGBoost and Random Forest as the two models in this project. They have both performed well on tabular trip data. Both generate feature importance outputs, ranking that indicates the input variables with the most significant contribution to the prediction (Ignatenko et al., 2024). This enables the researchers to not only know how accurate the model is, but also what features are most important in predicting demand in various zones.

2.2.3 SDG 11: Sustainable cities and communities and its applicability in predicting taxi demand.

2.2.3.1 The global view of SDG 11 and transport access.

One of the 17 goals established by the United Nations as part of the 2030 Agenda is SDG 11: Sustainable Cities and Communities. Target 11.2 is one of its targets; it urges all governments to make transport safe, affordable and accessible to everyone by 2030 with particular regard to vulnerable people. The indicators of how much progress toward this target have been achieved include Indicator 11.2.1, which monitors the proportion of the population with convenient access to public transport. Gathering data of 2,039 cities in 188 countries, the United Nations Department of Economic and Social Affairs (2024) claims that only 6 people in 10 cities in the world have easy access to public transport and the difference between more developed and less developed areas is significantly higher. This implies that despite the fact that some of the wealthiest cities in the world can still be found, a significant part of the population does not have access to public transport reliably. It is not merely an issue of construction of additional stations and routes. It is concerned with ensuring that the ones that are being served are the people who require transport the most. Cities such as New York City, with their outer borough residents who cannot choose as well and have worse access to personal cars are precisely the type of population that SDG 11 Target 11.2 is intended to safeguard.

2.2.3.2 How machine learning may help to achieve SDG 11.2.

Studies have already started to understand how machine learning models can be applied to not only enhance prediction accuracy but also to facilitate a more equitable transport planning. Yousefzadeh Barri et al. (2022) examined the performance of various machine learning algorithms in the context of predicting transit utilization among low-income inhabitants of the Greater Toronto and Hamilton Area, in particular. They discovered that machine learning models were more predictive of low-income riders than traditional statistical models, and demonstrated that the model selection may influence the appearance of the communities that seem to most require transit investment. This is a very important discovery to this project. In case the model that is predicting the demand of taxis is not effective in some areas, then those areas will not see the transport planners and their citizens remain under-served. This is directly tackled in the zone-stratified framework created in this

project which calculates individual accuracy scores of each of the 263 NYC taxi zones. This project gives the evidence that the planners must make decisions to support SDG 11 Target 11.2 on a community level by identifying areas where prediction quality fails and superimposing that information with transit accessibility data.

2.3 Similar Systems/ Works

Similar work 1: Correa and Moyano (2023) used more than 90 million trip records of New York City to predict taxi and Uber demand using both XGBoost and Random Forest. They employed 5 attributes: zone code, hour, day of week, temperature, and precipitation. XGBoost had an R-squared of 0.97 and Random Forest had 0.95. The study however did not provide zone-level scores but only aggregate scores. It took 263 zone dummy variables as input features, which could perhaps have enabled the model to memorise zone-level demand patterns instead of learning meaningful features. The research omitted transit accessibility or demographic characteristics and in its future research report, it stated it had not explored transit proximity in the outer boroughs of the city.

Similar work 2: Zhao et al. (2021) calculated theoretical predictability limits of the taxi demand on a building block level in Manhattan. They compared five prediction algorithms such as Neural Network, LSTM and a Markov predictor. They found that on average 83% of the maximum predictability was attained and that various building blocks possess radically different levels of predictability. Markov predictor performed better in high predictability regions and only consumed 0.02 percent of the computation time, as compared to LSTM. Nevertheless, the experiment was performed on Manhattan alone and was not able to generate real demand forecasts or performance by all 263 zones.

Similar work 3: Liu and Chen (2026) examined the for-hire vehicle records in New York City between 2019 and 2023 to examine how various neighbourhoods reacted to the COVID-19 pandemic disruption. They developed a two-layered framework on how to evaluate Urban Mobility Resilience. Two types of data were united in this structure. The former type was used to get changes in travel patterns of each neighbourhood over time. The second one outlined the socioeconomic and spatial attributes of every neighbourhood, including income and proximity to the city centre (Liu and Chen, 2026). Comparing these two layers, they classified neighbourhoods into three groups according to the patterns of their

travel response to the pandemic. The former exhibited Stability, i.e. travel demand remained about the same over the period. The second group revealed Recovery, i.e. the demand declined in the pandemic and then returned to normal. The third group was Adaptation meaning that the travel patterns changed permanently and never returned to the patterns before it. Their greatest discovery was that there was a distinct separation between the centre of the city and its periphery. More affluent neighbourhoods along the city centre remained unchanged. Their commuting needs were not so high since there were numerous means of transport accessible to the residents. The suburbs of the city led a different way. Rather than revert to pre-pandemic travel patterns, these regions adapted by changing the timing and mode of their travel, becoming more reliant on for-hire vehicles as their daily travel mode (Liu & Chen, 2026). The authors stated that their framework is reproducible and it was structured in such a way that allows policymakers to identify areas where transport inequality is being exacerbated, in accordance with SDG 11. But no prediction models were constructed in the study. It did not measure the capability of the machine learning algorithm to predict demand. It failed to generate zone-level accuracy scores. The research indicated the areas of inequality but was not trying to forecast future demand in such regions.

Similar work 4: Zhang et al. (2021) constructed a taxi demand prediction model in all 263 taxi zones in NYC. They referred to their model as Multi-Level Recurrent Neural Network or MLRNN. Recurrent neural network is a deep learning model that takes data in a single step sequence. It takes the data of the last step to aid in predicting the next step in every step (Mienye et al., 2024). One issue that Zhang et al. (2021) observed was training one model to cover the entire city. The patterns of demand differ drastically in different zones. A Manhattan area that is busy will not act like a quiet residential area in Staten Island. They referred to this issue zone as heterogeneity. To address this, they clustered zones together depending on the level of similarity in their patterns of demand with time. Two zones were put in the same group when their numbers of demand increased and decreased in the same hours and days. Zhang et al. (2021) referred to this as demand correlation. After grouping the zones, they trained a distinct recurrent neural network to each of the groups. In this manner, the only thing that each model needed to do is to learn the types of zones that acted similarly. It was actually a better predictive method than using only one model to predict the entire city (Zhang et al., 2021). Nevertheless, they used a deep learning model, but not an ensemble tree-based model. Grinsztajn et al. (2022) demonstrated that deep learning models do not perform as well as tree-based models on tabular data such as taxi trip records. No transit

accessibility or demographical features were also not included in the study. This implies that it was unable to reflect the local circumstances that influence the demand in outer boroughs areas where transportation is scarce.

Similar work 5: Alam et al. (2025) compared four machine learning models (NYC TLC data): Generalized linear regression, multinomial logistic regression, Random Forest, and Decision Tree. GLR had the best R-squared of about 90%. They discovered that tree-based models failed to work because of the skewed demand data, with the majority of the zones having low demand and a low number of zones having high demand. The analysis lacked any contextual/zone-level analysis.

Similar work 6: Hasnine et al. (2021) examined the factors that compel people to use ride-hailing apps such as Uber in Toronto, Canada. They utilized trip records, which were provided to them by a research agreement between them and the City of Toronto. These records were compared with the two kinds of information concerning each zone. The initial one was constructed environment data, and it explained the appearance of each zone. This encompassed the number of apartment buildings, types of land-use and employment in the area. The second one was weather data, which contained temperature and rainfall (Hasnine et al., 2021). They applied regression models to determine the most strongly correlated of these factors with the demand for ride-hailing. They found that either of the types of factors is significant. Areas that contained more apartments and had a higher ratio of land use had higher ride-hailing trips. Areas of single-family homes generated less trips. Weather was also a factor. In case of bad weather, the number of people who preferred ride-hailing over walking to a bus station or a subway increased (Hasnine et al., 2021). The importance of these findings to the present project is that they reveal that the physical layout of a zone contributes to understanding the number of ride-hailing trips that it produces. This justifies the inclusion of contextual variables such as proximity to the subway and the rate of car ownership in the predictive model. But Hasnine et al. (2021) failed to apply machine learning algorithms. They employed statistical regression, which tests the existence of a relationship between variables but does not give future demand forecasts. They have not given prediction accuracy scores such as R^2 or RMSE. They also failed to compare the performance of their model in various zones. The research indicated what factors were correlated with demand but failed to attempt to predict the appearance of the demand tomorrow or next week.

Similar work 7: Cats et al. (2022) examined Uber trip data in six cities and concluded that one-fifth to one-fourth of ride-hailing trips lacked a feasible mode of transit. Their analysis has shown that ride-hailing is a complement to transit in low-coverage regions, and a competitor to transit in high-coverage regions. The study was, however, observational and did not construct prediction models.

Table 1: Comparison Table of Similar Systems

Study	City/Data	Models	Features	Eval Level	Transit Feat.	Demo. Feat.	Key Finding	Gap This Project Fills
Correa & Moyano (2023)	NYC TLC 90M trips	XGBoost, RF	Zone code (263 dummies), hour, day, temp, rain	Aggregate only	No	No	$R^2=0.97$ XGBoost, $R^2=0.95$ RF	No zone-level eval, no contextual features
Zhao et al. (2021)	NYC Manhattan blocks	NN, LSTM, Markov, ARIMA, LZW	Temporal + weather (wind speed)	Building block	No	No	Max predictability 83%, varies by block	Manhattan only, theoretical bounds not actual forecasts
Liu & Chen (2026)	NYC FHV 2019–2023	None (descriptive resilience framework)	Mobility dynamics + neighbourhood context	Neighbourhood level	No	Yes	Centre-periphery divide in mobility resilience	No prediction model, no zone-level accuracy
Zhang et al. (2021)	NYC 263 zones	MLRNN (deep learning)	Temporal + zone cluster	Cluster level	No	No	Zone clustering improves DL accuracy	Deep learning only, no contextual features

Alam et al. (2025)	NYC TLC	GLR, MNL, RF, DT	Trip features	Aggregate only	No	No	GLR best at $R^2 \approx 90\%$, RF underperformed due to overfitting	No zone-level eval, no transit features
Hasnine et al. (2021)	Toronto TNC	None (statistical regression)	Weather + built environment (apartments, land use diversity)	Zone level	No	Partial (built env.)	Both weather and the built environment affect TNC demand	No ML prediction model, no forecast accuracy
Cats et al. (2022)	6 cities Uber	None (descriptive)	Trip + transit data	City level	Yes	No	20–40% trips have no viable PT alternative	No prediction model, observational only

2.3.2 Summary of Gaps

The comparison shows that there are three gaps that are uniform in all the seven studies. First, no research determines the performance of the models at individual zone level in all 263 NYC taxi zones. Only aggregate scores were reported in Correa and Moyano (2023) and Alam et al. (2025). Zhang et al. (2021) performed zone clustering, but did not generate per-zone statistics. Second, none of the studies includes both the features of transit accessibility and demographics in a demand prediction model. The study by Hasnine et al. (2021) analyzed the impacts of weather and built environment but failed to construct prediction models. Third, there is no research that links performance of zone-level models to transit accessibility evidence to determine underserved communities. Liu and Chen (2026) found spatial inequality in travel demand but could not associate it with the accuracy of

prediction. The present project tackles the three gaps by integrating zone-stratified assessment with contextual aspects and generating an interactive dashboard that links the predictive performance with transit accessibility.

2.4 Technical Research

In this section, the tools and technologies to be utilized in this project are reviewed. Each technology is explained in terms of what it does, the reason why it was chosen and its application in the project.

2.4.1 Python

Python is a general-purpose programming language and is popular in data science and machine learning (Sarker, 2021). It was chosen in this project due to the Correa and Moyano (2023) anchor study, which processed and trained their models using Python. Python offers a huge ecosystem of data manipulation, machine learning and visualisation libraries that is why it is appropriate to a project that involves data cleaning, feature engineering, model training, and dashboard development.

2.4.2 Jupyter Notebook

Jupyter Notebook is an open-source web app that allows users to write and execute Python code in small units known as cells (Kluyver et al., 2016). A cell may contain code, text or a chart. A single cell can be operated or multiple cells can be operated simultaneously. This allows checking the output at every work step easily. Kluyver et al. (2016) defined Jupyter Notebook as a type of publication of reproducible computational workflow, i.e. it enables researchers to provide their code together with the output of these computations. This project makes use of Jupyter Notebook to perform all data exploration, preprocessing, and model trainings since it enables the researcher to visualize the data after every cleaning step before proceeding to the next step.

2.4.3 Pandas

Pandas is a Python package to manipulate table-based data (McKinney, 2010). It lists information in rows and columns in a format known as a DataFrame which resembles a spreadsheet. Pandas are used to perform all the data cleaning, merging and feature engineering in this project. These involve eliminating invalid trip records, extracting

time-related aspects of date columns such as hour of the day, and merging the trip data with the weather and census data using common key columns. The library was selected as Correa and Moyano (2023) applied the same types of operations to their data with the help of pandas and keeping the methods similar helps to maintain the methods.

2.4.4 NumPy

NumPy is a Python library to work with numbers and arrays (Harris et al., 2020). It offers rapid mathematical computations that are all used by pandas, scikit-learn, and XGBoost under the hood. Harris et al. (2020) defined NumPy as the basic array computing package in Python. NumPy is employed in doing numerical operations in this project like the straight-line distance between the locations of the zone centres and the subway stations.

2.4.5 Scikit-learn

Scikit-learn is a Python machine learning library that offers an implementation of popular algorithms such as Random Forest, and model evaluation algorithms such as R squared and RMSE (Pedregosa et al., 2011). This project applies it to train the Random Forest model, chronologically splits the data, and calculates zone-level assessment measures using scikit-learn. scikit-learn was also applied in the anchor study by Correa and Moyano (2023).

2.4.6 Random Forest (implementation with scikit-learn)

The Random Forest implementation in this project is the RandomForestRegressor class of scikit-learn (Pedregosa et al., 2011). It trains a collection of decision trees that are independently built with bootstrap sampling and makes predictions by averaging the decisions of all the trees. The controlled comparison with the enriched model is maintained with default hyperparameters as explained in Section 2.2.4.3.

2.4.7 XGBoost

The XGBoost is an open-source library that uses a gradient boosted decision tree algorithm (Chen & Guestrin, 2016). It constructs trees one after another, with each tree fixing the mistakes of the last tree. According to Correa and Moyano (2023), XGBoost is a user-friendly version of gradient tree boosting formalised to control overfitting. In the project, the XGBRegressor class of the xgboost Python library is used with default hyperparameters.

2.4.8 GeoPandas

GeoPandas is a Python package which extends pandas with geographic data operations (Jordahl et al., 2021). It allows the researcher to load the map files, merge datasets according to the geographic location, and calculate the distance between points on a map. In this project, GeoPandas is implemented to read the shapefile of taxi zones on NYC, locate the centre point of each zone, and determine the distance between each zone centre and the closest subway station of the MTA.

2.4.9 Plotly

Plotly is a Python library to create interactive charts and maps (Plotly Technologies Inc., 2015). Contrary to the static chart libraries, Plotly enables users to hover, zoom on and filter data points in real time. In this project, Plotly is applied to create the interactive dashboard. The dashboard provides zone-wise R^2 , RMSE, subway closeness, and car ownership rate of the 263 taxi zones in NYC. Each zone can be clicked on by the user to view all the results in that zone.

2.4.10 Matplotlib

Matplotlib is a python library for creating static charts and graphs (Hunter, 2007). It is applied in this project in the exploratory data analysis during the preprocessing stage. This involves plotting demand distribution at the borough level, demand distribution by hour of the day, and testing whether the features are correlated with each other. Matplotlib was developed by Hunter (2007) to create figures that can be easily understood in scholarly articles.

2.4.11 Operating System

The project is created and implemented on a local computer with a regular operating system and Python 3.x installed. There is no cloud computing infrastructure adopted. The entire processing, model training and dashboard generation are done on-site which limits the amount of data that can be processed but assures complete reproducibility without external dependencies.

2.5 Summary

2.5.1 Domain Research overview.

The field study examined the literature in four major fields that are pertinent to this project. The former domain determined that taxi and for-hire vehicles have a dual role in urban transport by acting as the connectors in areas with low coverage of transit and competing with the transit in well-served areas. Within the city of New York, in particular, 86 percent of all taxi pickups are centered in Manhattan with the outer boroughs being underserved. The second field studied machine learning methods to predict taxi demand and discovered that ensemble tree based models like XGBoost and random forest are better than deep learning models on tabular trip data because they are resilient to uninformative features, can learn features with independent meaning, and learn irregular patterns. The third area appraised is engineering practices and discovered that temporal features are not effective in predicting demand in low demand areas where the number of trip records is limited. Research proved that the addition of contextual characteristics like proximity to transit and sociodemographic conditions can enhance prediction accuracy as compared to temporal characteristics. The fourth area dealt with model assessment and discovered that total performance ratings hide considerable disparities in model performance in individual zones. The evaluation at the zone level is required to detect where models fail, and default hyperparameters are used to give a clean experimental baseline to compare feature sets.

2.5.2 Overview of Related Literature.

Seven studies were contrasted in nine dimensions in order to determine gaps in current literature. Three gaps were identified in the comparison. To begin with, there is no prior study that compares the performance of models at the individual zone level of all 263 taxi zones in NYC. Second, none of the studies considers both the access to transit and demographics in a demand prediction model. Third, none of the studies links the performance of zone-level models with evidence of transit accessibility in order to identify underserved communities. The three gaps are fulfilled in the present project with zone-stratified evaluation framework, feature-rich, and interactive dashboard.

2.5.3 Overview of Technical Research.

The project implements scikit-learn with Python as a programming language, the implementation of the Random Forest and the implementation of the gradient boosted trees as the XGBoost library. Pandas and NumPy deal with data manipulation. GeoPandas is used to carry out spatial operations like calculating the proximity of subway stations. The interactive dashboard has been developed with the help of Plotly. All the tools are open-source and were chosen to be compatible with the anchor study by Correa and Moyano (2023) and to guarantee complete reproducibility of the findings.

CHAPTER 3: METHODOLOGY

3.1 Introduction

The chapter explains the way the project was planned, the way the data was collected and the way in which it was prepared in readiness to get to the modelling stage. It will start by presenting three data analytics methodologies, their comparison and the justification of the final selection. It then records the 5 datasets that will be used in this project, their sources, their contents and why they were chosen. The chapter proceeds to the preprocessing procedures used to clean and merge the datasets, and finally, the exploration of the final dataset using visualizations and statistical summaries. It ends with a conclusion about the main findings of the methodology choice as well as the data preparation work.

3.2 Methodology

3.2.1 Introduction of methodology

Data analytics projects are concerned with large amounts of raw data that must be collected, cleaned, explored, modelled, and ultimately transformed into a valuable outcome. All of this would be hard to do without a guiding framework so that it could be easy to arrange the work and make it repeatable and in the correct sequence. A data analytics methodology offers such a framework. It provides a sequence of stages during which the researcher proceeds through the project up to the conclusion, and each stage is based on the outcomes of the previous one (Schroer et al., 2021). In the literature, three data analytics methodologies are typically mentioned: CRISP-DM, KDD, and SEMMA. All of them have somewhat different approaches to the way in which a data mining project is to be organized.

CRISP-DM is an acronym that means Cross Industry Standard Process for Data Mining, developed in 2000 by a group of industry partners, such as Daimler Chrysler, SPSS, and NCR (Plotnikova et al., 2020). It has six steps: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment. The phases are presented as a cycle, which implies that the researcher will be able to shift to a previous phase at any time in case discoveries necessitate it. Schroer et al. (2021) defined CRISP-DM as the

de facto standard in data mining projects, verifying in a systematic review of 24 published papers in IEEE, ScienceDirect and ACM that it is the most popular framework used in research and practice. It is characterized by the fact that it starts with the Business Understanding stage, which requires the researcher to explicitly state the problem and the goals of the project before any data is manipulated.

KDD or Knowledge Discovery in Databases was initially defined by Fayyad et al. (1996) as a general process of knowledge discovery in data. It comprises nine steps that go through comprehension of the application domain up to finding and interpreting patterns. In contrast to CRISP-DM, KDD is linear-based (the steps follow each other in a predetermined order) and does not have an explicit means of going back to a previous step (Plotnikova et al., 2020). The KDD lacks a deployment phase as well. The process concludes with the interpretation of the patterns discovered, without giving a formal instruction to the researcher of how to transform the patterns into a product or tool to be used by the end users.

SEMMA (Sample, Explore, Modify, Model, and Assess) is a tool created by SAS Institute as an extension of their SAS Enterprise Miner software (Plotnikova et al., 2020). It is divided into five steps, beginning with a sample of the data, to the evaluation of the model performance. Plotnikova et al. (2020) reported that SEMMA is a vendor-specific process model, i. e. it was created to be used in a specific commercial software environment, not as a general one. They also noted that SEMMA does not follow two steps that KDD considers crucial: namely, understanding the application domain at the beginning, and integrating the learned knowledge back into the system at the end. This implies that SEMMA does not have a Business Understanding phase, nor does it have a Deployment phase.

3.2.2 Justification and choice of methodology.

Having introduced all three methodologies, it is time to compare them and choose the one that will be suitable for the project. The comparison is made in seven criteria given in Table 3.1 below.

Table 2: Comparison of CRISP-DM, KDD, and SEMMA

Criteria	CRISP-DM	KDD	SEMMA
Full name	Cross-Industry Standard Process for Data Mining	Knowledge Discovery in Databases	Sample, Explore, Modify, Model, Assess
Number of phases	6	9	5
Process direction	Iterative. Allows returning to earlier phases (Plotnikova et al., 2020)	Linear. Follows a fixed sequence (Plotnikova et al., 2020)	Linear. Follows a fixed sequence (Plotnikova et al., 2020)
Business Understanding phase	Yes. The very first phase defines the problem and objectives	No. Begins directly with data selection	No. Begins with data sampling
Deployment phase	Yes. The final phase delivers the output to end users	No. Ends at interpretation of patterns	No. Ends at model assessment
Tool independence	Yes. Works with any programming language or platform	Yes. Not tied to a specific vendor	No. Built for SAS Enterprise Miner (Plotnikova et al., 2020)
Research adoption	De-facto standard in data mining (Schröer et al.,	Widely referenced but limited modern adoption	Rarely used outside SAS environments (Plotnikova et al.,

	2021)	(Plotnikova et al., 2020)	2020)
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The comparison indicates evident differences. CRISP-DM only starts with Business Understanding and finishes with Deployment. KDD ends with interpretation and does not provide any advice on how to present the results, whereas SEMMA does not define any problems or deploy them, which is identified as one of the significant limitations by Plotnikova et al. (2020).

The second significant dissimilarity is the direction of the process. CRISP-DM is constructed in a cycle. In case, during data preparation, something arises that alters how the researcher perceives the problem, the framework officially permits to backtrack to the Business Understanding or Data Understanding phase to modify. This is the iterative execution that Plotnikova et al. (2020) defined as the most notable difference between CRISP-DM and KDD as the latter presupposes that the researcher proceeds to the next step of the nine steps and does not go back to the beginning.

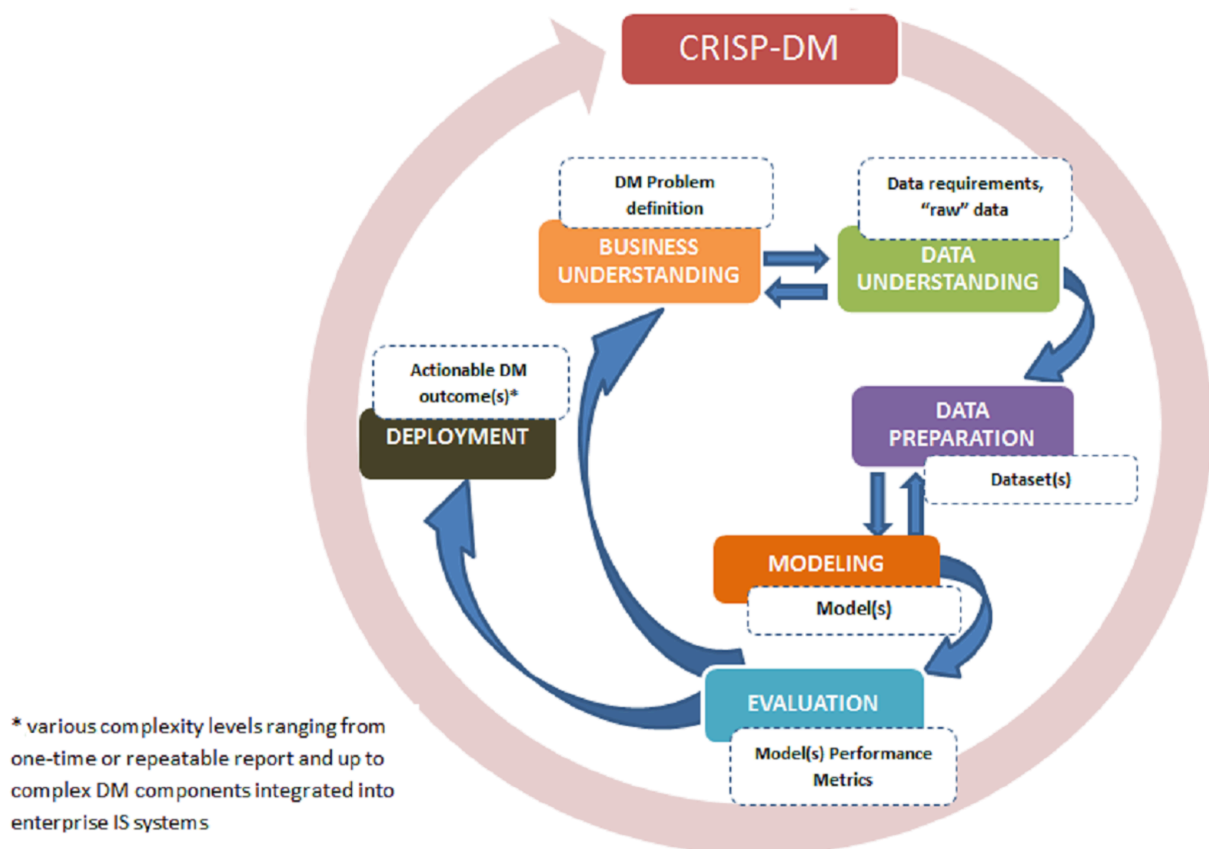
The third is tool independence. CRISP-DM and KDD can both be used with any software or programming language. SEMMA is not. SAS Institute created it as a component of their SAS Enterprise Miner product and Plotnikova et al. (2020) referred to it as a vendor-specific model, which restricts its use in other settings. All this project is completely based on Python open-source libraries pandas, scikit-learn, XGBoost, GeoPandas, and Plotly. All these are not SAS tools.

CRISP-DM is chosen as the methodology for this project. It is the only framework that requires the researcher to state the problem and provides a practical output at the end - which is required here as the project starts with three clearly-defined problem statements regarding the transport inequality in NYC and ends with an interactive dashboard to transport planners. It is iterative in design which implies that the project is able to re-visit previous stages when there are unexpected problems in pre-processing. And its tool independence implies that it is friendly with the Python ecosystem that is the foundation of this project. CRISP-DM is still the most popular data mining framework in published research according to the systematic review of Schröer et al. (2021), which further validates its decision.

3.2.3 Phases of CRISP-DM applied to this project

The six stages of CRISP-DM are laid out in an iterative cycle. The standard process diagram is shown in Figure 2 below.

Figure 2: CRISP-DM circular phase diagram showing all six phases



Note. From "Adaptations of data mining methodologies: a systematic literature review," by Plotnikova et al., 2020, *PeerJ Computer Science*, 6, e267 (<https://doi.org/10.7717/peerj-cs.267>). Copyright 2020 by Plotnikova et al. Reprinted with permission.

Phase 1: Business Understanding (Semester 1, Weeks 1-4)

The initial step of CRISP-DM is Business Understanding. It concerns itself with knowing what the project is attempting to achieve through the eyes of the individuals who will utilize the outcomes. (Schröer et al., 2021). The researcher specifies the problem, determines the goals, the target users and what a successful result would look like. This was

defined by Schroer et al. (2021) as the step in which the research objectives are transformed into data mining problem definitions.

This project expressed three problem statements that were based on gaps identified in the available literature. The first is that none of the studies has tested the accuracy of machine learning models per zone (out of 263 taxi zones in NYC). The second is that the current models lack contextual characteristics, like the proximity of subways or the car ownership rates (Correa and Moyano, 2023; Maharjan et al., 2024). The third is that there is no tool that can integrate zone-level model performance with evidence of transit accessibility to transport planners (Liu and Chen, 2026). Based on the three issues, a project purpose, three quantifiable goals, two target audiences, and a SDG 11 Target 11.2 (United Nations, 2015) mapping were developed. All this is written in Chapter 1.

Phase 2: Data Understanding (Semester 1, Weeks 36)

Data Understanding is a stage where the datasets required by the project are identified, collected, and explored initially (Schroer et al., 2021). The researcher will go through the structure of each dataset, explore the columns it has, identify any apparent quality problems, and will start to form a mental image of what the data looks like. This initial exploration is sometimes referred to as exploratory data analysis, and is intended to guide the subsequent decisions that follow - what should be cleaned, which features may be useful, and whether the data is actually answering the research questions.

In this project, there were five open-source government datasets that were gathered in this phase. All datasets were loaded, examined in terms of column names, data type, date, and value distributions, and verified against missing or unforeseen values. This early experimentation directly influenced the preprocessing of Phase 3, as it showed the datasets that required the most cleaning, and what transformations would be required to unite them. In Section 3.3, the datasets are described and in Section 3.4.2, the exploratory visualisations are presented.

Phase 3: Data preparation (Semester 1, Weeks 510)

The Data Preparation stage involves the cleaning, transformation and integration of raw data into a machine learnable form (Schroer et al., 2021). Schroer et al. (2021) determined that this stage existed in each of the studies included in their systematic review, which proves that no real-world dataset is ready to be modelled. The particular actions undertaken in this stage are motivated by what was found in Data Understanding the problems that the exploratory analysis uncovered are the problems that preprocessing will solve. The cleaning of each of the five datasets in this project was done according to what was observed during the preliminary inspection.

The TLC data showed invalid trip records, which were eliminated once the exploration had shown that there were null pickup zone IDs, a value of 0 passengers and a negative fare. The pickup datetime column was analyzed to obtain three temporal features: hour of day, day of week and month. The individual rows were summed up into zone-hour records. All 263 zones were calculated with GeoPandas based on the proximity of the subway stations. Aggregation of census tract-level data to the zone level was done through area-weighted averaging of car ownership. NOAA Central Park station temperature data were combined with the zone-hour data by matching by date and hour. The result is a single zone-hour DataFrame that has all the features required in the baseline and in the enriched models. In Section 3.4.1, the entire preprocessing pipeline is reported.

Phase 4: Modelling (Semester 2)

The step of training machine learning algorithms on the ready dataset, as well as their application to predict, is called modelling (Schroer et al., 2021). The researcher chooses the algorithms, sets up their parameters, trains them on the training data, and makes predictions on the test data.

Two ensemble tree-based models, XGBoost and Random Forest, are trained twice in this project, once with the baseline feature set of temporal features and temperature, and once with the enriched feature set that includes the proximity of subway stations and the rate of car ownership. An 80/20 chronological train-test split is used, which is consistent with that of Correa and Moyano (2023). Throughout, default hyperparameters are used. Bischl et al. (2023) demonstrated that hyperparameters could significantly impact model performance hence, holding them fixed in all four model runs would mean that variations in accuracy are due to the features and not the tuning.

Phase 5: Evaluation (Semester 2)

Evaluation is the step at which the trained models are evaluated on the data that the models have not encountered throughout training (Schroer et al., 2021). The researcher evaluates the models against the success criteria set during the Business Understanding phase and finds out the areas where performance is good and where it is not.

In this project, assessment is more than a normal one aggregate score. R-squared and RMSE are calculated on each of the 263 NYC taxi zones, and the zone-stratified evaluation framework of Chapter 1 is generated. As demonstrated by Meyer and Pebesma (2022), a spatial model may have high overall accuracy and fail in particular underrepresented areas. These scores at the zone level are added together with the proximity of the subway and car ownership data to indicate zones with a poor score in model accuracy and poor transit access. Such areas are referred to as underserved communities.

Phase 6: Deployment (Semester 2)

The stage where the results are transformed into a product accessible and actionable by the target users is the stage of deployment (Schroer et al., 2021). According to Schroer et al. (2021), deployment is the least-implemented step of published CRISP-DM research, with 17 of 24 studies in their review not having implemented it at all.

Deployment in this project will be in the form of an interactive dashboard constructed with Plotly. The dashboard shows R-squared, RMSE, proximity to the subway and car ownership rate in each of the 263 zones on a single screen. Areas with low model accuracy and transit access are visually identified. Liu and Chen (2026) showed the importance of developing a reproducible framework that can assist policymakers in detecting socio-spatial inequalities. In this project, the dashboard is useful to the transport planners in New York City.

3.3 Data Collection

Five open-source government datasets were collected for this project. All of them contribute to at least one of the three goals of Chapter 1. Each of them is freely available on the Internet and may be downloaded without restrictions on use in research. In Briefing 3, Miss Amira stressed that CSDA students should make sure that their datasets are not illegal to use. All of the five datasets of this project are official government data portals.

3.3.1. Trip records: The Yellow Taxi TLC in NYC.

Source: NYC Taxi and Limousine Commission Open Data Portal

Link: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

Format: Parquet files, one per month

Period: January to March 2025

Key columns: tpep_pickup_datetime, PULocationID, passenger_count, trip_distance, fare_amount (NYC TLC, 2025)

This is the main source of data. The row corresponds to one taxi trip and the column that is the most important is PULocationID, a column that corresponds to each trip being assigned a specific one of the 263 official NYC taxi zones. After the aggregation of the data to the zone-hour level, the data is the target variable and this is the number of pickups that the models are trained to predict.

The yellow taxi data was selected instead of the green taxi data by the fact that the research question is a specific question of the Manhattan concentration problem. Correa and Moyano (2023) discovered that in NYC, yellow taxis make up about 86 percent of every taxi pickup, and that the large majority of these are in Manhattan. The outer boroughs were to be served by green taxis (Xie et al., 2021), so their inclusion would minimize the very imbalance which the project is meant to explore.

The data is 2025 as opposed to the previous period Correa and Moyano (2023) utilized. Liu and Chen (2026) demonstrated that the COVID-19 pandemic permanently altered the travel patterns within NYC - outer borough neighbourhoods had to adapt, altering how and when people travelled, and the changes have not been undone. Alam et al. (2025)

affirmed that the NYC TLC fleet did not resume to approximately 302,000 vehicles until 2024, which structurally was not the same as it was before the pandemic. The 2025 data training is based on the way the city moves today, which the model learns.

This data is openly published under NYC Open Data licence without restrictions.

3.3.2 Taxi zone shapefile and lookup table in NYC.

Source: NYC Taxi and Limousine Commission Open Data Portal

Link: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page> (under Taxi Zone Maps and Lookup Tables)

Format: Five shapefile component files (.shp, .shx, .dbf, .prj, .cpg) and a CSV lookup table (NYC TLC, 2025)

Coverage: All 263 official NYC taxi zones across five boroughs

The shapefile contains the polygon boundary of each taxi zone. The new table is used to look up the names and boroughs of each zone ID. They work together to give the spatial support to all the zone-based calculations in the project.

Spatial joins between the MTA station data and the Census data to the appropriate zones are done using the zone polygons. The centroid of each polygon is used to compute subway station proximity. The zone IDs are both the joining key in all five datasets and the grouping variable when assessing on the zone level. This is an open dataset without limitations.

3.3.3 MTA subway station data.

Source: New York State Open Data Portal, Metropolitan Transportation Authority

Link: <https://data.ny.gov/Transportation/MTA-Subway-Stations/39hk-dx4f>

Format: CSV

Key columns: Stop Name, Borough, GTFS Latitude, GTFS Longitude (MTA, 2024)

In this dataset, the geographical positions of all subway stations and Staten Island Railway stations in New York City can be found. It is applied to calculate subway station proximity - the straight-line distance between the centre of each taxi zone to the closest MTA station.

One of the two contextual features in the enriched model is the proximity to the subway. In their future work section, Correa and Moyano (2023) admitted that the question-whether the outer borough taxi pickups can cluster around subway stations or not was never considered. Wang et al. (2023) determined that distance to transit is among five dimensions of built environment which influence taxi demand patterns within a city. With the help of this dataset, it is possible to test the hypothesis whether the introduction of the subway proximity as a feature will enhance the predictive accuracy in the outer boroughs.

This data is accessible on the New York State Open Data portal and is not restricted.

3.3.4 American community survey table B08141 US Census.

Source: United States Census Bureau, American Community Survey 5-Year Estimates, 2020– 2024

Link for CSV: <https://data.census.gov/table/ACSDT5Y2024.B08141>

Link for ShapeFile:

Format: CSV at census tract level

Key columns: Geographic ID, total households, zero-vehicle households (U.S. Census Bureau, 2024)

American Community Survey is an ongoing national survey administered by the US Census Bureau which gathers data regarding the household living, working and traveling (U.S. Census Bureau, 2024). The 5-Year Estimate is a combination of five years of responses and is therefore more reliable with small regions such as census tracts where one year would not yield a big sample.

Table B08141 captures the number of households per census tract and the number owning no vehicle; and the division of the two provides the zero-vehicle rate. Maharjan et al. (2024) have discovered that high zero-vehicle areas have the poorest transit coverage - the residents of such areas are the most taxi-dependent. This condition is not observed in Correa and Moyano (2023) locally.

There is no overlap between census tracts and taxi zone boundaries. One taxi zone may even cross portions of multiple tracts. To deal with this, area-weighted averaging is employed. The car ownership rates per zone are calculated as a weighted average of all the overlapping tracts. A tract larger than a zone receives a large weight. An arc which cuts a little corner receives a small one.

The US Census Bureau has no restrictions on the release of this data publicly.

3.3.5 NOAA integrated surface database -weather data.

Source: National Oceanic and Atmospheric Administration, Integrated Surface Database

Link:<https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>

Direct file: <https://www.ncei.noaa.gov/data/global-hourly/access/2025/72505394728.csv>

Station: Central Park, New York City (WBAN: 94728)

Format: CSV with hourly rows

Period: January to March 2025

Column used: Temperature in degrees Fahrenheit only

Correa and Moyano (2023) identified the third most predictive feature of taxi demand using feature importance analysis, temperature, following hour of day and day of week. Their statistics revealed that their demand is higher when the weather is below 36 degrees Fahrenheit as no one wants to wait in the open air to take the transport. On a freezing morning and a warm afternoon, the model would predict the same demand.

Temperature is only extracted out of this data set. Precipitation is not possible since Correa and Moyano (2023) found that rainfall did not have a significant predictive value. Wind speed, which was Zhao et al.(2021) was part of a larger weather feature pool of their Manhattan study, which is also excluded to maintain the feature set of the baseline in the anchor study to make a clean comparison.

This is publicly available, free of restrictions, and with no restrictions, in the form of data at NOAA.

3.4 Initial Data Pre-processing and Data Understanding

3.4.1 Dataset 1: NYC TLC Yellow Taxi Trip Records

Step 1: Loading and Merging the Files

The pandas and glob libraries were used to load the three monthly Parquet files (January to March 2025) and combine them into a single DataFrame. There were 11,198,026 rows and 20 columns in the raw merged dataset.

```

Total raw rows loaded (Jan-Mar): 11,198,026
Total columns in dataset: 20

--- Column Names and Data Types ---
VendorID      int32
tpep_pickup_datetime  datetime64[us]
tpep_dropoff_datetime  datetime64[us]
passenger_count  float64
trip_distance    float64
RatecodeID      float64
store_and_fwd_flag  object
PULocationID    int32
DOLocationID    int32
payment_type     int64
fare_amount      float64
extra            float64
mta_tax         float64
tip_amount       float64
tolls_amount     float64
improvement_surcharge  float64
total_amount     float64
congestion_surcharge  float64
Airport_fee      float64
cbd_congestion_fee  float64
dtype: object

```

Figure 3: Validated Merged Data

Step 2: Date Range Filtering

A check on the pickup datetime column showed that there were problems with the quality of data. The latest trip date was recorded in 2007-12-05 and the oldest had an extension beyond March 2025 to April 2025. These were records that were not within the study window and were eliminated. The dataset was narrowed down to 11,198,001 rows after only keeping the trips between 1 January 2025 and 31 March 2025.

```
cbd_congestion_fee      float64
# 1. Verify the exact Start and End dates
min_date : Any = df_tlc['tpep_pickup_datetime'].min()
max_date : Any = df_tlc['tpep_pickup_datetime'].max()

print(f"Earliest trip in dataset: {min_date}")
print(f"Latest trip in dataset: {max_date}")
print("-" * 50)
```

21 ✓ A0.1s Pacific University of Technology and Innovation

```
Earliest trip in dataset: 2007-12-05 18:45:00
Latest trip in dataset:  2025-04-01 00:00:17
-----
```

Figure 4: Error in Timeline

Getting the Corrected Range

```

start_date = '2025-01-01'
end_date = '2025-03-31 23:59:59'

Step 2: Date Range Filtering
# Keep only the rows that fall exactly within our 3-month window
df_clean: DataFrame = df_tlc[(df_tlc['tpep_pickup_datetime'] >=
start_date) & ("tpep": Unknown word.datetime').min()
max_date) & (df_tlc['tpep_pickup_datetime'] <= end_date)]

min_date: Any = df_clean['tpep_pickup_datetime'].min()
max_date: Any = df_clean['tpep_pickup_datetime'].max()
print("-" * 50)

print(f"Earliest trip in dataset: {min_date}")
print(f"Latest trip in dataset: {max_date}")
print("-" * 50)

print(f"Row count after date filtering: {len(df_clean):,}")

```

163] ✓ 5.1s

```

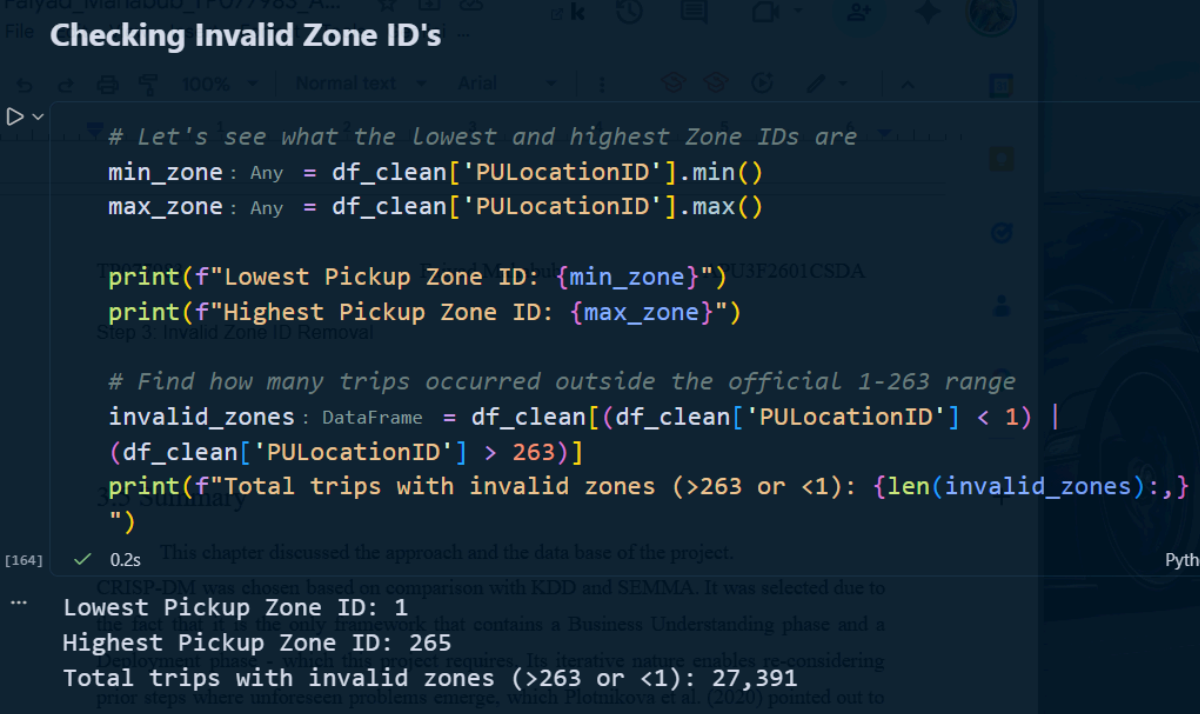
... Earliest trip in dataset: 2025-01-01 00:00:00
Latest trip in dataset: 2025-03-31 23:59:59
-----
3.5 Summary
Row count after date filtering: 11,198,001

```

Figure 5: Filtering the corrected data out from Jan to March

Step 3: Invalid Zone ID Removal

There are 263 valid zone IDs of 1 to 263 which are defined by the official NYC TLC taxi zone system. A review of PULocationID column revealed that the zone IDs were between 1 and 265, indicating that certain trips had zone codes which do not match with any of the official taxi zones. The number of trips that had invalid zone IDs was found to be 27,391 and eliminated. The result of this step was a dataset with 11,170,610 rows.



```

File Checking Invalid Zone ID's
# Let's see what the lowest and highest Zone IDs are
min_zone: Any = df_clean['PULocationID'].min()
max_zone: Any = df_clean['PULocationID'].max()

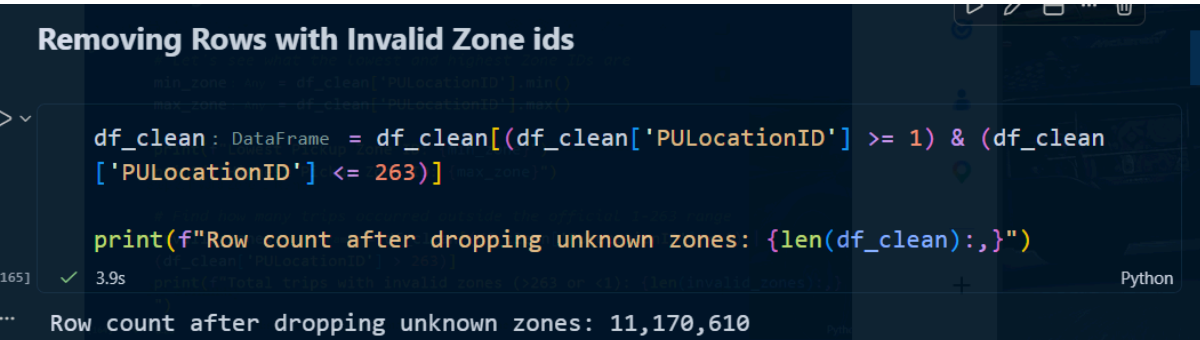
print(f"Lowest Pickup Zone ID: {min_zone}")
print(f"Highest Pickup Zone ID: {max_zone}")

# Find how many trips occurred outside the official 1-263 range
invalid_zones: DataFrame = df_clean[(df_clean['PULocationID'] < 1) |
(df_clean['PULocationID'] > 263)]
print(f"Total trips with invalid zones (>263 or <1): {len(invalid_zones):,}")

[164] ✓ 0.2s This chapter discussed the approach and the data base of the project.
CRISP-DM was chosen based on comparison with KDD and SEMMA. It was selected due to
the fact that it is the only framework that contains a Business Understanding phase and a
...
Lowest Pickup Zone ID: 1
Highest Pickup Zone ID: 265
Total trips with invalid zones (>263 or <1): 27,391

```

Figure 6: Invalid Zones are checked



```

Removing Rows with Invalid Zone ids
df_clean: DataFrame = df_clean[(df_clean['PULocationID'] >= 1) & (df_clean
['PULocationID'] <= 263)]

print(f"Row count after dropping unknown zones: {len(df_clean):,}")

[165] ✓ 3.9s
Row count after dropping unknown zones: 11,170,610

```

Figure 7: Removal of Invalid Zone ID

Step 4: Outlier Removal

The fare amount and trip distance columns were checked on outliers. The findings revealed some extreme values that were evidently errors in the data like the highest fare of 863372.12 and the highest distance of the trip 320136 miles. Also, 539,327 journeys had zero or negative fares and 290,759 journeys had a zero trip distance. All these records were destroyed. A limit of 500 was used on fares and a limit of 100 miles to trip distance was used in order to eliminate the extreme values and keep the valid long-distance airport trips. The dataset had 10,389,410 rows after removal of outliers.

```

... Trips with $0 or negative fares: 539,327
Trips with 0.0 mile distance: 290,759

--- Summary Statistics (Looking for extreme maximums) ---
      fare_amount  trip_distance
count  11170610.00    11170610.00
mean      17.22         6.18
std       262.30        582.45
min      -1807.60         0.00
25%         8.60         1.00
50%        12.80         1.72
75%        20.50         3.24
max      863372.12    320136.29

```

Figure 8: Detection of outliers

```

print("Fixing the Outliers Problem...")

# 1. Filter out <= 0 fares and distances
df_clean: DataFrame = df_clean[(df_clean['fare_amount'] > 0) & (df_clean
['trip_distance'] > 0)]

# 2. Filter out extreme maximums (Capping fare at $500 and distance at 100
miles) or Removal
df_clean: DataFrame = df_clean[(df_clean['fare_amount'] <= 500) & (df_clean
['trip_distance'] <= 100)]

print(f"Row count after dropping outliers: {len(df_clean):,}")

[167] ✓ 6.7s Python

... Fixing the Outliers Problem...
Row count after dropping outliers: 10,389,410

```

Figure 9: Removal of outliers

Step 5: Ghost Trip Removal

Trips that had zero or null passenger were termed as ghost trips, that is, trips that were captured in the system but had no passenger. The number of trips with a null number of passengers was 1,767,347 and the one with zero number of passengers was 65,595. Both types were eliminated. The dataset left after this step had 8,556,468 valid trips, which were 82.36 of the post-outlier dataset.

```
... -----  
Ghost Trips (NULL passenger_count): 1,767,347  
Ghost Trips (ZERO passenger_count): 65,595  
-----  
Valid Journeys (> 0 passengers): 8,556,468  
3.5 Summary  
Percentage of Valid Data: 82.36%  
This chapter discussed the approach and the data base of the project.
```

Figure 10: Identification of ghost rider

```
... Row count BEFORE removing ghost trips: 10,389,410  
Row count AFTER removing ghost trips: 8,556,468
```

Figure 11: Identification of ghost rider

Step 6: Missing Values Check

A null value test was conducted after all the cleaning processes to check all columns. The outcome of the result affirmed the absence of any null values in any column of the cleaned data.

```

Edit View Insert Format Tools Gemini ...
--- Null Value Count per Column ---
VendorID 0
tpep_pickup_datetime 0
tpep_dropoff_datetime 0
passenger_count 0
trip_distance 0
RatecodeID 0
store_and_fwd_flag 0
PULocationID 0
DOLocationID 0
payment_type 0
fare_amount 0
extra 0
mta_tax 0
tip_amount 0
tolls_amount 0
improvement_surcharge 0
total_amount 0
congestion_surcharge 0
Airport_fee 0
cbd_congestion_fee 0
dtype: int64

```

Figure 12: No Missing Values

Step 7: Column Selection

The cleaned dataset only retained the two columns that were needed to build the model. These were tpep pickup datetime column and PULocationID column. The final cleaned data had 8,556,468 rows that represent 257 different pickup areas of January to March 2025.

```
... Final valid rows: 8,556,468
Date Range: 2025-01-01 00:00:00 to 2025-03-31 23:59:59
Total Zones: 257
--- Column Names and Data Types ---
tpep_pickup_datetime datetime64[us]
PULocationID int32
dtype: object
```

Figure 13: Final Column Selected

Step 8: Temporal Feature Extraction

The pickup datetime column had four temporal features. These were the date, hour of day (0 to 23), day of week (0 to 6, Monday to Sunday) and month (1 to 3). These characteristics constitute the feature set at the training of the model.

```
# 1. Extract the 'date' for grouping
df_final['date'] = df_final['tpep_pickup_datetime'].dt.date

# 2. Extract the Hour of Day (0 to 23)
df_final['hour'] = df_final['tpep_pickup_datetime'].dt.hour

# 3. Extract the Day of Week (0=Monday to 6=Sunday)
df_final['day_of_week'] = df_final['tpep_pickup_datetime'].dt.dayofweek
Date Range: 2025-01-01 00:00:00 to 2025-01-31 23:59:59

# 4. Extract the Month (1 to 3 for your Jan-Mar window)
df_final['month'] = df_final['tpep_pickup_datetime'].dt.month

# Validation: Check the new features
print(df_final[['tpep_pickup_datetime', 'hour', 'day_of_week', 'month']].head())
```

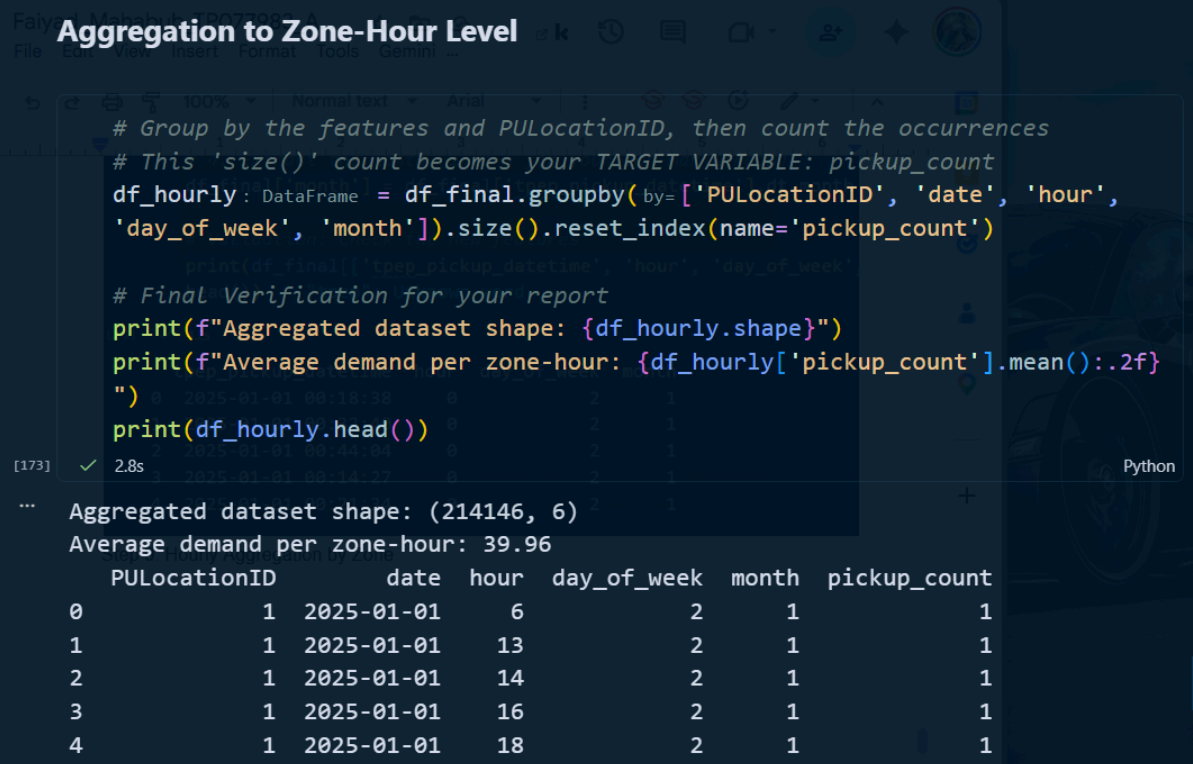
[172] ✓ 4.2s

	tpep_pickup_datetime	hour	day_of_week	month
0	2025-01-01 00:18:38	0	2	1
1	2025-01-01 00:32:40	0	2	1
2	2025-01-01 00:44:04	0	2	1
3	2025-01-01 00:14:27	0	2	1
4	2025-01-01 00:21:34	0	2	1

Figure 14: Extraction of Temporal Features

Step 9: Hourly Aggregation by Zone

The trip-level data were summarized to the zone-hour level with PULocationID, date, hour, day of week and month grouped and the frequency of each group being counted. This number was made the target variable known as pickup_count. The consolidated data had 214,146 rows with a mean demand of 39.96 pickups per zone-hour.



```

# Group by the features and PULocationID, then count the occurrences
# This 'size()' count becomes your TARGET VARIABLE: pickup_count
df_hourly: DataFrame = df_final.groupby(by=['PULocationID', 'date', 'hour',
      'day_of_week', 'month']).size().reset_index(name='pickup_count')

print(df_final[['trip_pickup_datetime', 'hour', 'day_of_week'])

# Final Verification for your report
print(f"Aggregated dataset shape: {df_hourly.shape}")
print(f"Average demand per zone-hour: {df_hourly['pickup_count'].mean():.2f}")
print(df_hourly.head())

```

[173] ✓ 2.8s Python

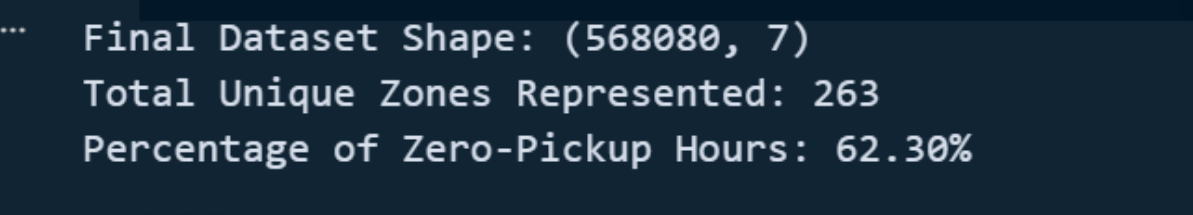
```

... Aggregated dataset shape: (214146, 6)
Average demand per zone-hour: 39.96

```

	PULocationID	date	hour	day_of_week	month	pickup_count
0	1	2025-01-01	6	2	1	1
1	1	2025-01-01	13	2	1	1
2	1	2025-01-01	14	2	1	1
3	1	2025-01-01	16	2	1	1
4	1	2025-01-01	18	2	1	1

Figure 15: Aggregation of zone by hourly



```

... Final Dataset Shape: (568080, 7)
Total Unique Zones Represented: 263
Percentage of Zero-Pickup Hours: 62.30%

```

Figure 16: Zero padding for missing zone hours

3.4.2 Dataset 2: MTA Subway Station

```

Total subway stations found: 496

--- Column Names ---
['GTFS Stop ID', 'Station ID', 'Complex ID', 'Division', 'Line', 'Stop Name', 'Borough', 'CBD', 'Daytime Routes', 'Str

--- Raw Data Sample ---

```

	Station ID	Complex ID	Stop Name	GTFS Latitude	GTFS Longitude
0	1	1	Astoria-Ditmars Blvd	40.775036	-73.912034
1	2	2	Astoria Blvd	40.770258	-73.917843
2	3	3	30 Av	40.766779	-73.921479
3	4	4	Broadway	40.761820	-73.925508
4	5	5	36 Av	40.756804	-73.929575

Figure 17: Raw Data Inspection

Pandas.read_csv was used to initially load the MTA Subway Station dataset. A preliminary search showed that there are 496 total subway stations in NYC with 19 columns that consist of: Station ID, Stop Name, Borough, GTFS Latitude and GTFS Longitude. A preview with a .head() confirmed that the structure was clean and readable on the first look.

Missing Values

The two most important columns (GTFS Longitude and GTFS Latitude) were specifically subjected to a null check. The outcome indicated that there were no missing values in either column, or that no imputation or dropping of rows were required when dealing with the coordinate data. The data was full to make the spatial analysis.

```

Missing Latitudes: 0
Missing Longitudes: 0

```

Figure 18: Missing Value Inspection

Outliers

--- Coordinate Range Check ---		
	GTFS Latitude	GTFS Longitude
count	496.000000	496.000000
mean	40.723205	-73.945948
std	0.082440	0.069813
min	40.512764	-74.251961
25%	40.672557	-73.986957
50%	40.715021	-73.950960
75%	40.773485	-73.903092
max	40.903125	-73.755405

Figure 19: Missing Value Inspection

In order to check the integrity of the coordinates, a check of descriptive statistics was performed on both columns of coordinates. The findings showed that all values were within the projected NYC bounding box.

Data Transformation

The cleaned subway DataFrame was transformed from a flat table into a GeoDataFrame using geopandas. Latitude and Longitude columns were converted into Shapely Point geometry objects. This transformation is essential to enable spatial operations such as distance calculations and spatial joins, which are not possible with raw numeric coordinates.

```

import geopandas as gpd
from shapely.geometry import Point

# Create the GeoDataFrame
# We use EPSG:4326 because raw Lat/Lon is always in degrees
subway_gdf = gpd.GeoDataFrame(
    df_subway_clean,
    geometry=gpd.points_from_xy(df_subway_clean['GTFS Longitude'], df_subway_clean['GTFS Latitude']),
    crs="EPSG:4326"
)

subway_gdf

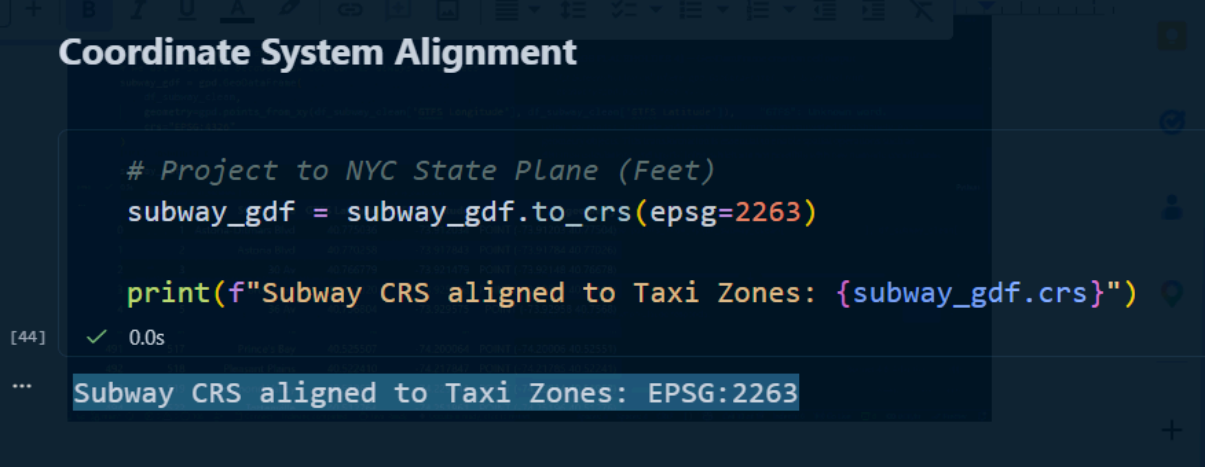
```

	Station ID	Stop Name	GTFS Latitude	GTFS Longitude	geometry
0	1	Astoria-Ditmars Blvd	40.775036	-73.912034	POINT (-73.91203 40.77504)
1	2	Astoria Blvd	40.770258	-73.917843	POINT (-73.91784 40.77026)
2	3	30 Av	40.766779	-73.921479	POINT (-73.92148 40.76678)
3	4	Broadway	40.761820	-73.925508	POINT (-73.92551 40.76182)
4	5	36 Av	40.756804	-73.929575	POINT (-73.92958 40.7568)
...	...	...	...	...	...
491	517	Prince's Bay	40.525507	-74.200064	POINT (-74.20006 40.52551)
492	518	Pleasant Plains	40.522410	-74.217847	POINT (-74.21785 40.52241)
493	519	Richmond Valley	40.519631	-74.229141	POINT (-74.22914 40.51963)
494	520	Tottenville	40.517764	-74.251961	POINT (-74.25196 40.51776)

Figure 20: Data Transformation

Data Standardization

In order to make the subway dataset and the taxi zone dataset consistent, they were both offset to the same Coordinate Reference System (CRS) - EPSG: 2263 (NYC State Plane, which is measured in feet). In the absence of this step, the distance calculations would be useless as the two datasets would not be in the same units. The subway GeoDataFrame was reprojected with the help of the `.to_crs(eps=2263)` method.



```
Coordinate System Alignment

# Project to NYC State Plane (Feet)
subway_gdf = subway_gdf.to_crs(eps=2263)

print(f"Subway CRS aligned to Taxi Zones: {subway_gdf.crs}")
```

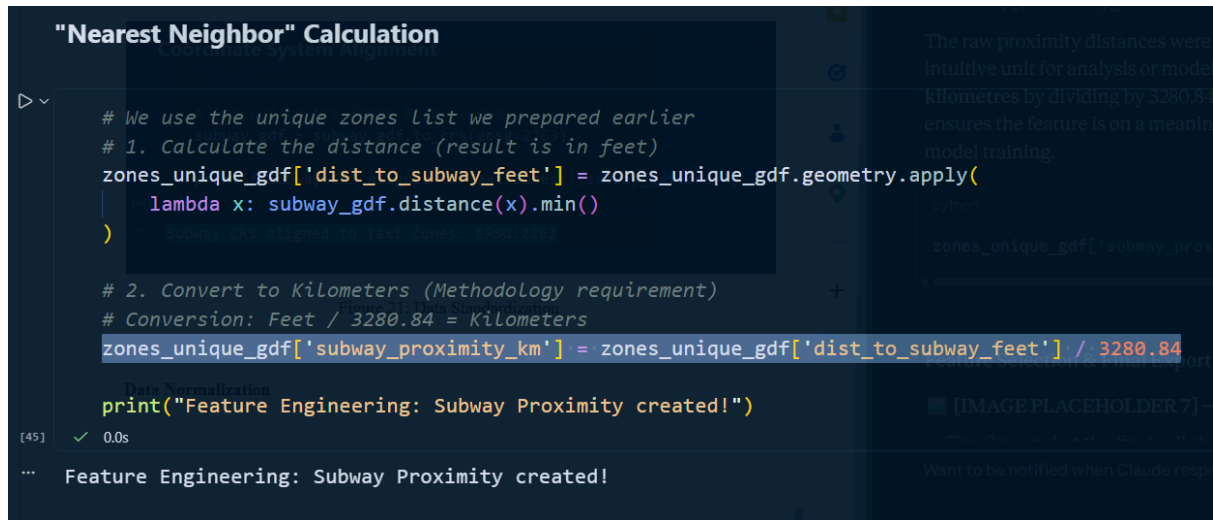
[44] ✓ 0.0s

... Subway CRS aligned to Taxi Zones: EPSG:2263

Figure 21: Data Standardization

Data Normalization

Raw proximity distances were calculated in feet (because of EPSG:2263), which is not an intuitive unit to analyse or model. These values were also normalized/converted to kilometres by dividing by 3280.84 to give the final subway_proximity_km feature. This makes sure that the feature is meaningfully and internationally standard on a scale to train downstream models.



```
"Nearest Neighbor" Calculation
Coordinate System Alignment

# We use the unique zones list we prepared earlier
# 1. Calculate the distance (result is in feet)
zones_unique_gdf['dist_to_subway_feet'] = zones_unique_gdf.geometry.apply(
    lambda x: subway_gdf.distance(x).min()
)
# Subway CRS aligned to Taxi Zones - EPSG:2263

# 2. Convert to Kilometers (Methodology requirement)
# Conversion: Feet / 3280.84 = Kilometers
zones_unique_gdf['subway_proximity_km'] = zones_unique_gdf['dist_to_subway_feet'] / 3280.84

print("Feature Engineering: Subway Proximity created!")

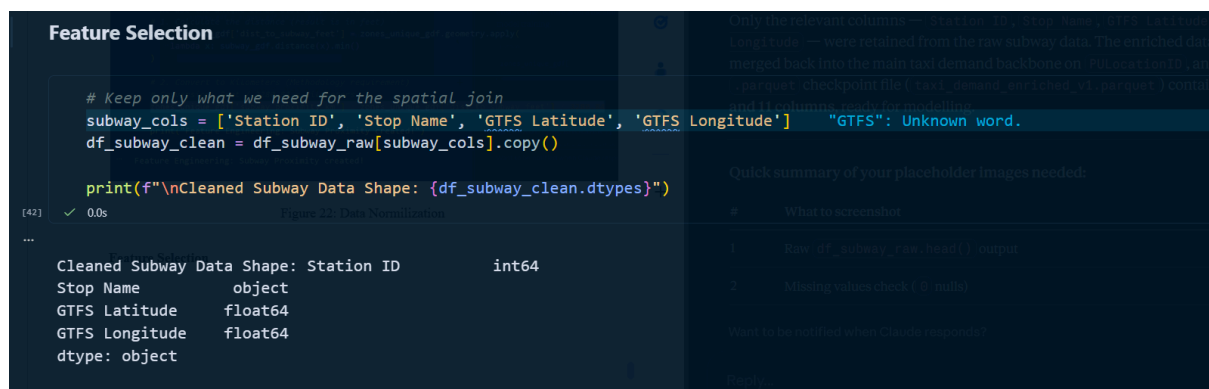
[45] ✓ 0.0s

... Feature Engineering: Subway Proximity created!
```

Figure 22: Data Normalization

Feature Selection

The raw subway data was only saved in the form of the relevant columns: Station ID, Stop Name, GTFS Latitude, and GTFS Longitude. The enriched dataset was once again inserted back into the original taxi demand backbone on PULocationID, and then exported as a .parquet checkpoint file (taxi_demand_enriched_v1.parquet) with 568,080 rows and 11 columns, which could be easily modelled.



The screenshot shows a Jupyter Notebook interface with a dark theme. The main code cell is titled "Feature Selection" and contains the following Python code:

```
# Keep only what we need for the spatial join
subway_cols = ['Station ID', 'Stop Name', 'GTFS Latitude', 'GTFS Longitude']
df_subway_clean = df_subway_raw[subway_cols].copy()

print(f"\nCleaned Subway Data Shape: {df_subway_clean.dtypes}")
```

The output of the code is displayed below the cell:

```
Cleaned Subway Data Shape: Station ID      int64
Stop Name      object
GTFS Latitude   float64
GTFS Longitude  float64
dtype: object
```

On the right side of the notebook, there is a sidebar with a search bar and a list of files. The top file is "taxi_demand_enriched_v1.parquet". Below it, there is a section titled "Quick summary of your placeholder images needed:" with a list of items: "What to screenshot", "Row df_subway_raw.head(1) output", and "Missing values check (is null)".

Figure 23: Feature Selection

3.5 Summary

This chapter discussed the approach and the data base of the project. CRISP-DM was chosen based on comparison with KDD and SEMMA. It was selected due to the fact that it is the only framework that contains a Business Understanding phase and a Deployment phase - which this project requires. Its iterative nature enables re-considering prior steps where unforeseen problems emerge, which Plotnikova et al. (2020) pointed out to be the most distinctive feature of it compared to the linear structure of KDD. It is also independent of tools, unlike SEMMA that was created on SAS Enterprise Miner (Plotnikova et al., 2020). Schroer et al. (2021) have verified that it has become the de-facto standard in data mining. Semester 1 is where the first three phases, namely Business Understanding, Data Understanding, and Data Preparation are done. Semester 2 is followed by Modelling, Evaluation, and Deployment.

Five open-source government datasets are gathered, cleaned and merged into one zonehour dataset. The preprocessing filtered out invalid trip records, derived temporal

characteristics, derived subway proximity (all 263 zones), and aggregated car ownership rates (census tracts to zones) using area-weighted averaging and combined hourly temperature data. The data understanding visualisations affirmed that Manhattan leads in the number of pickups (proving Problem Statement 1 and the necessity of zone-stratified analysis), that the patterns of hourly and daily demand are robust and repeatable (satisfying the baseline feature set), and that there are initial patterns among subway proximity, car ownership rate, and zone-level demand (proving Objective 2). These results form the basis of Semester 2.

CHAPTER 4: CONCLUSION

4.1 Discuss the achievement of the first part of the project

The preliminary part of this project has been accomplished successfully in all the four chapters. The introduction chapter made it very clear that the problem of taxi demand prediction models in New York City had never been tested with a specific consideration to New York City. This conceals the fact that the models may not be performing very well in places such as Staten Island or the Bronx. Three problem formulations were composed to determine what was lacking in existing studies. These included the absence of zone-by-zone model testing, absence of transit and car ownership data in available models and the unavailability of a practical planning tool to the transport authorities.

The literature review discussed the operation of taxis in a city and the use of machine learning to forecast taxi demand. It also described the reason why tree based models such as XGBoost and Random Forest are more appropriate to this kind of data than deep learning models. Seven related studies were evaluated and compared and three obvious gaps in research were revealed that will be addressed in the project.

The methodology chapter had been done as well. CRISP-DM methodology was chosen as the most suitable measure to use in this project of data analytics and was justified as compared to KDD and SEMMA. There were 5 open-source government datasets. These were the NYC TLC Yellow Taxi Trip Records in January to March 2025, the NYC TLC Taxi Zone Shapefile, MTA Subway Station data, NOAA hourly weather data and the US Census ACS Table B08141 of car ownership rates. All the datasets are publicly available and found on official government portals. In the Jupyter Notebook, the process of data preprocessing was done using Python. This included the management of missing data, deleting invalid trip data and deriving time-related data of hour of the day and day of the week. GeoPandas was used to conduct spatial joins between every taxi zone and its closest distance to a subway station and car ownership rate. Initial data understanding was also accomplished. The important variables were identified and described, observations through the five boroughs were compared and histograms were generated to realize the distribution of taxi demand according to zones and different periods of time of the day.

The project plan was finished within the timeframes and all the tasks were submitted on 22 April 2026.

4.2 Justify whether the student was able to do enough investigation/research concerning what to achieve for the project. Ensure relate your investigation results with the selected SDG goal.

This project was well-grounded enough due to adequate research conducted. The study affirmed that Manhattan is receiving approximately 86% of the total taxi pickups with the outer boroughs such as Staten Island and Queens being underserved. Researchers Correa and Moyano (2023) and Zhao et al. (2021) have studied in detail and discovered that neither of these researches examined the performance of their models in each individual zone. This clearly justifies the necessity of the project.

This inquiry resonates well with SDG 11: Sustainable Cities and Communities, Target 11.2, which stipulates that all people and more so vulnerable communities should have affordable and accessible transportation. The residents of the outer borough, who use taxis but are seldom studied, are precisely the population that SDG 11.2 targets to safeguard. This project can provide city planners with the evidence they require to allocate more taxi resources to those areas by identifying areas where the model is performing poorly and where access to transit is limited.

4.3 Justify if there are any gaps in the research and design areas that the student may want to further explore and improve.

Although the investigation stage was comprehensive, there were some areas that were not exhaustively covered and will be covered in Semester 2. The actual model training and testing is the most important task that is left. The datasets are gathered and preprocessed but not yet built and analyzed at the zone level using XGBoost and Random Forest models. This is the main activity of Semester 2.

Hyperparameter tuning was also not included on the current plan to make sure that any performance difference between the baseline and the enriched models is only due to the

additional features. The enriched model can be investigated using a simple grid search in Semester 2 assuming they have time. Lastly, deep learning systems like LSTM were not included in this project to limit the comparison and enable a one-to-one comparison with the anchor study by Correa and Moyano (2023). Future research on this project can extend the framework to deep learning models.

REFERENCES

- Aboueela, M., Stamos, I., Chaniotakis, M. E., & Antoniou, C. (2025). Disentangling the relationships between the SDGs and transportation. *Research in Transportation Business & Management*, 62, Article 101447. <https://doi.org/10.1016/j.rtbm.2025.101447>
- Alam, S., Ayub, M. S., Cui, H., & Khan, M. A. (2025). A comparative study of machine learning models for taxi-demand prediction using a big data framework. *Public Transport*, 17, 803–833. <https://doi.org/10.1007/s12469-025-00401-1>
- Barajas, J. M., & Brown, A. (2021). Not minding the gap: Does ride-hailing serve transit deserts? *Journal of Transport Geography*, 90, 102918. <https://doi.org/10.1016/j.jtrangeo.2020.102918>
- Bischl, B., Binder, M., Lang, M., Pielok, T., Richter, J., Coors, S., Thomas, J., Ullmann, T., Becker, M., Boulesteix, A.-L., Deng, D., & Lindauer, M. (2023). Hyperparameter optimization: Foundations, algorithms, best practices, and open challenges. *WIREs Data Mining and Knowledge Discovery*, 13(2), e1484. <https://doi.org/10.1002/widm.1484>
- Cats, O., Kucharski, R., Danda, S. R., & Yap, M. (2022). Beyond the dichotomy: How ride-hailing competes with and complements public transport. *PLOS ONE*, 17(1), e0262496. <https://doi.org/10.1371/journal.pone.0262496>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>
- Chicco, D., Warrens, M. J., & Jurman, G. (2021). The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation. *PeerJ Computer Science*, 7, e623. <https://doi.org/10.7717/peerj-cs.623>
- Chou, K. S., Wong, K. L., Zhang, B., Aguiari, D., Im, S. K., Lam, C. T., Tse, R., Tang, S. K., & Pau, G. (2023). Taxi demand and fare prediction with hybrid models: Enhancing efficiency and user experience in city transportation. *Applied Sciences*, 13(18), 10192. <https://doi.org/10.3390/app131810192>
- Correa, D., & Moyano, C. (2023). Analysis and prediction of New York City taxi and Uber demands. *Journal of Applied Research and Technology*, 21(5), 886–898. <https://doi.org/10.22201/icat.24486736e.2023.21.5.2074>
- Correa, D., & Moyano, C. (2022). Dynamics of the growth of Uber vs green taxis in outer boroughs in New York City. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4229008>

- Evelly, J. (2024, March 15). More New Yorkers are struggling to afford public transit: Report. *City Limits*.
<https://citylimits.org/more-new-yorkers-are-struggling-to-afford-public-transit-report/>
- Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). From data mining to knowledge discovery in databases. *AI Magazine*, 17(3), 37–54.
- Flor, M., Ortuño, A., & Guirao, B. (2022). Ride-hailing services: Competition or complement to public transport to reduce accident rates. *Frontiers in Future Transportation*, 3, 935937. <https://doi.org/10.3389/ffutr.2022.935937>
- Gangrade, A., Pratyush, P., & Hajela, G. (2022). Taxi-demand forecasting using dynamic spatiotemporal analysis. *ETRI Journal*, 44(4), 624–640.
<https://doi.org/10.4218/etrij.2021-0123>
- Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*. <https://doi.org/10.48550/arXiv.2207.08815>
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerevelen, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362.
<https://doi.org/10.1038/s41586-020-2649-2>
- Hasnine, M. S., Hawkins, J., & Habib, K. N. (2021). Effects of built environment and weather on demands for transportation network company trips. *Transportation Research Part A: Policy and Practice*, 150, 171–185. <https://doi.org/10.1016/j.tra.2021.06.005>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- Ignatenko, V., Koren, V., Kuznetsov, A., & Burnaev, E. (2024). Tree-based algorithms for tabular data: A practical comparison. *PeerJ Computer Science*, 10, e1775.
<https://doi.org/10.7717/peerj-cs.1775>
- Ignatenko, V., Surkov, A., & Koltcov, S. (2024). Random forests with parametric entropy-based information gains for classification and regression problems. *PeerJ Computer Science*, 10, e1775. <https://doi.org/10.7717/peerj-cs.1775>
- ISI. (2024). Optimizing urban mobility: A comparative analysis of taxi demand prediction models. *Ingénierie des Systèmes d'Information*, 29(5). <https://doi.org/10.18280/isi.290522>
- Jin, S. T., Kong, H., & Sui, D. Z. (2019). Uber, public transit, and urban transportation equity: A case study in New York City. *The Professional Geographer*, 71(2), 315–330.
<https://doi.org/10.1080/00330124.2018.1531038>

- Jordahl, K., Van den Bossche, J., Fleischmann, M., Wasserman, J., McBride, J., Gerard, J., Tratner, J., Perry, M., Garcia Badaracco, A., Farmer, C., Hjelle, G. A., Snow, A. D., Cochran, M., Gillies, S., Culbertson, L., Barber, M., Ward, B., Caria, G., Taves, M., ... Leblanc, A. (2021). *geopandas/geopandas: v0.9.0*. Zenodo.
<https://doi.org/10.5281/zenodo.4569086>
- Kluyver, T., Ragan-Kelley, B., Pérez, F., Granger, B., Bussonnier, M., Frederic, J., Kelley, K., Hamrick, J., Grout, J., Corlay, S., Ivanov, P., Avila, D., Abdalla, S., & Willing, C. (2016). Jupyter Notebooks — a publishing format for reproducible computational workflows. In F. Loizides & B. Schmidt (Eds.), *Positioning and Power in Academic Publishing: Players, Agents and Agendas* (pp. 87–90). IOS Press. doi:
https://ui.adsabs.harvard.edu/link_gateway/2016ppap.book...87K/doi:10.3233/978-1-61499-649-1-87
- Li, P., Chang, J., Zhang, Y., & Zhang, Y. (2021). Multi-zone prediction analysis of city-scale travel order demand. *PLOS ONE*, 16(3), e0248064.
<https://doi.org/10.1371/journal.pone.0248064>
- Liu, Y., & Chen, M. (2026). An open framework for assessing urban mobility resilience: Evidence from New York City. *npj Urban Sustainability*, 6, 62. .
<https://doi.org/10.1038/s42949-026-00368-3>
- Maharjan, S., Janatabadi, F., & Ermagun, A. (2024). Spatial inequity of transit and automobile access gap across America for underserved population. *Transportation Research Record*, 2678(1), 674–690. <https://doi.org/10.1177/03611981231171914>
- Maharjan, S., Tung, L. W., & Venigalla, M. M. (2024). Examining public transit accessibility to jobs: Modal vs. multimodal analyses. *Transportation Research Record*, 2678(1), 674–690. <https://doi.org/10.1177/03611981231171914>
- McKinney, W. (2010). Data structures for statistical computing in Python. In S. van der Walt & J. Millman (Eds.), *Proceedings of the 9th Python in Science Conference* (pp. 56–61).
<https://doi.org/10.25080/Majora-92bf1922-00a>
- Meyer, H., & Pebesma, E. (2022). Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nature Communications*, 13(1), 2208.
<https://doi.org/10.1038/s41467-022-29838-9>
- Mienye, I. D., Swart, T. G., & Obaido, G. (2024). Recurrent neural networks: A comprehensive review of architectures, variants, and applications. *Information*, 15(9), 517.
<https://doi.org/10.3390/info15090517>
- MTA. (2024). New York City Transit. <https://www.mta.info/agency/new-york-city-transit>

- MTA. (2024). MTA Subway Stations [Data set]. New York State Open Data Portal.
<https://data.ny.gov/Transportation/MTA-Subway-Stations/39hk-dx4f>
- New York City Department of City Planning. (2025). *New York City's population estimates and trends: May 2025 release*.
https://www.nyc.gov/assets/planning/downloads/pdf/our-work/reports/new-york-city-population-estimates-and-trends_may-2025.pdf
- NYC TLC. (2025). TLC trip record data. NYC Taxi and Limousine Commission.
<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- New York City Taxi and Limousine Commission. (2025). *February 2025 for-hire vehicle license review report and determination*.
<https://www.nyc.gov/assets/tlc/downloads/pdf/license-pause-report-2025-02.pdf>
- Office of the New York City Comptroller. (2025). *Behind schedule: How New York City's bus system slow rolls riders*.
<https://comptroller.nyc.gov/reports/behind-schedule-how-new-york-citys-bus-system-slow-rolls-riders/>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
<https://jmlr.org/papers/v12/pedregosa11a.html>
- Peng, D., Huang, M., & Xing, Z. (2023). Taxi origin and destination demand prediction based on deep learning: A review. *Digital Transportation and Safety*, 2(3), 176–189.
<https://doi.org/10.48130/DTS-2023-0014>
- Plotly Technologies Inc. (2015). Collaborative data science. Plotly Technologies Inc.
<https://plot.ly>
- Plotnikova, V., Dumas, M., & Milani, F. (2020). Adaptations of data mining methodologies: A systematic literature review. *PeerJ Computer Science*, 6, e267.
<https://doi.org/10.7717/peerj-cs.267>
- Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. *SN Computer Science*, 2(3), 160. <https://doi.org/10.1007/s42979-021-00592-x>
- Schröer, C., Kruse, F., & Gómez, J. M. (2021). A systematic literature review on applying CRISPDM process model. *Procedia Computer Science*, 181, 526–534.
<https://doi.org/10.1016/j.procs.2021.01.199>
- U.S. Census Bureau. (2024). American Community Survey 5-Year Estimates, Table B08141.
<https://data.census.gov/table/ACS5Y2024.B08141>

- United Nations Department of Economic and Social Affairs. (2024). Goal 11: Make cities inclusive, safe, resilient and sustainable. <https://sdgs.un.org/goals/goal11>
- United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development (A/RES/70/1). <https://sdgs.un.org/2030agenda>
- US Census Bureau. (2023, May 1). *American National Standards Institute (ANSI), Federal Information Processing Series (FIPS), and other standardized geographic codes*. Census.gov. <https://www.census.gov/library/reference/code-lists/ansi.html#cousub>
- Wang, Y., Duan, Y., Mao, X., Zhao, J., & Ma, N. (2023). Influence of the built environment on taxi travel demand based on the optimal spatial analysis unit. PLOS ONE, 18(10), e0292363. <https://doi.org/10.1371/journal.pone.0292363>
- Xie, Y., Zhao, K., Sun, Y., & Chen, D. (2021). Revealing spatiotemporal travel demand and community structure characteristics with taxi trip data: A case study of New York City. PLOS ONE, 16(11), e0259694. <https://doi.org/10.1371/journal.pone.0259694>
- Yousefzadeh Barri, E., Farber, S., Jahanshahi, H., & Beyazit, E. (2022). Understanding transit ridership in an equity context through a comparison of statistical and machine learning algorithms. Journal of Transport Geography, 105, Article 103482. <https://doi.org/10.1016/j.jtrangeo.2022.103482>
- Zhang, C., Zhu, F., Lv, Y., Ye, P., & Wang, F. (2021). MLRNN: Taxi demand prediction based on multi-level deep learning and regional heterogeneity analysis. IEEE Transactions on Intelligent Transportation Systems, 23(7), 8412–8422. <https://doi.org/10.1109/TITS.2021.3080511>
- Zhao, K., Khryashchev, D., & Vo, H. (2021). Predicting taxi and Uber demand in cities: Approaching the limit of predictability. IEEE Transactions on Knowledge and Data Engineering, 33(6), 2723–2736. <https://doi.org/10.1109/TKDE.2019.2955686>

APPENDICES

Appendix A: PPF

Project Title Proposal		
Project Title: Evaluating Machine Learning Prediction Performance Across New York City Taxi Zones to Identify Underserved Communities		Status: APPROVED
Proposed By	FAIYAD MAHABUB TASIN	
Description	Introduction New York City runs one of the largest urban transportation systems in the world, in which more than 195,487 vehicles are for-hire vehicles, i.e. those vehicles licensed by the New York City Taxi and Limousine Commission (TLC) where they operate to carry passengers at a fare, serving millions of residents in five geographically separate boroughs. Nevertheless, there are severe inequalities in transportation across those boroughs. The Metropolitan Transportation Authority (MTA) subway system is clustering the stations closer to the center of 5.3 in Manhattan and 0.7 in Queens and none in Staten Island (MTA, 2024). Such spatial inequality compels residents of outer boroughs to rely on cab and ride-hailing apps to travel to their destinations in a disproportionate manner. Precise and location-specific taxi demand forecasting is a valuable instrument in ensuring equitable allocation of transport planning. Machine learning (ML) refers to algorithms trained to identify patterns in large-scale data to help in making predictions about previously unseen and unseen observations. Correa and Moyano (2022) used two ensemble tree-based models, i.e., XGBoost (Extreme Gradient Boosting) and Random Forest, on more than 90 million records of trips in New York City and found the aggregate R-squared (R^2) scores of 0.97 and 0.95 respectively, where shown how closely model predictions match actual values on a scale from 0 to 1. But these scores have been calculated at the aggregate level and were never disaggregated by zone. The aggregate measure is very much biased toward the highly served, high-demand areas of Manhattan since about 80% of all taxi pickups in their data were in Manhattan and tells them nothing about the actual performance of the model in the outer boroughs. This exact issue was exhibited by Mayer and Peabson (2022), who revealed that a model that claims to have a good overall accuracy score may still fail in a systematic way in areas that are underrepresented in training data. A second limitation compounds this problem. Neither Correa and Moyano (2022) nor Zhao et al. (2021) included contextual features which are input variables that describe the local conditions of a geographic zone such as subway station proximity or car ownership rate in their prediction models. Mahajan et al. (2024) demonstrated that households without a vehicle are disproportionately clustered in suburban and rural areas where transit coverage is weaker, confirming that these variables capture structurally meaningful differences between zones. Despite this evidence, Correa and Moyano (2022) acknowledged in their future work section that whether outer borough taxi pickups cluster near subway stations was never examined in their study. A third gap follows directly from the first two. Even if zone-level evidence of model failure were available alongside contextual transit data, no existing study has combined these two sources into a single tool that transport planners can act on. Liu and Chen (2024) demonstrated the value of connecting mobility data with neighborhood context by producing a reproducible framework that identified socio-spatial transport inequalities across New York City in support of Sustainable Development Goal (SDG) 11. No equivalent tool exists for taxi demand prediction. This project therefore addresses all three gaps by applying XGBoost and Random Forest within a zone-stratified evaluation framework across all 263 New York City taxi zones, incorporating contextual features including subway station proximity and car ownership rate into an enriched prediction model, and producing an interactive dashboard that integrates zone-level prediction performance with transit accessibility indicators. The project is aligned with SDG 11: Sustainable Cities and Communities, specifically Target 11.2, which calls for access to safe, affordable, and sustainable transport systems for all, with special attention to the needs of those in vulnerable situations (United Nations, 2015). Project Aim To evaluate machine learning model prediction performance at the individual taxi zone levels across all five boroughs of New York City using XGBoost and Random Forest with contextual transit and demographic data, to assist transport planners in determining which communities require increased taxi service. Objectives To evaluate the zone-level prediction performance of XGBoost and Random Forest models trained on temporal and weather features across all 263 New York City taxi zones, computing individual Root Mean Squared Error (RMSE) and R-squared (R^2) scores for each zone as a baseline for comparison. To assess whether incorporating contextual features, such as subway station proximity and car ownership rate, improves model prediction accuracy in low-demand outer borough zones compared to the baseline model. To produce an interactive dashboard that integrates zone-level prediction performance metrics from the enriched model alongside transit accessibility indicators, to help transport planners identify which communities across all five boroughs require prioritized taxi service deployment. Target Users Urban transport planners and city authorities in New York City who require zone-level demand forecasts and transit accessibility evidence to make data-driven decisions about prioritizing taxi resource deployment across underserved outer borough communities. Data science researchers in the urban mobility domain who may reuse the merged dataset or the zone-stratified evaluation framework to extend this work to other modal types, time periods, or cities. SDG SDG 11 Keywords Machine Learning Analytics and Data Web Scraping Python Performance Analytics Preferred Supervisor(s) ASSOC. PROF. DR. NIKASH FATHIMA ABUBACKER, DR. PREETHI SUBRAMANIAN, DR. KUAN YIE JIANN, DR. ADGLINE SNEHA JOHN CHRISASTUM, TS. DR. LAH FOONG LI Assigned Supervisor nfeetha.abubacker Assigned Second Marker rahman	
Remarks		
Name	Remarks	Date
NUR AMIRA BINTI ABDUL MAJID	Good	Mar 12, 2026, 12:21:45 AM

Figure 24: Project Proposal

Appendix B: Ethics Forms (Fast Track)

Ethics Form

FAIYAD MAHABUB TASIN

Supervisor Remarks

Name	Remarks	Date
ASSOC. PROF. DR. NIRASE FATHIMA ABUBACKER	Acceptable	Apr 17, 2026, 4:34:52 AM

Ethics Form (Fast-Track)

1Participant Confidentiality

2Nature of Research

3Target Participants

4Support Information

Participant Confidentiality

Will you describe the main procedures to participants in advance, so that they are informed about what to expect?

☐ Yes

☐ No

☒ N/A

Will you tell participants that their participation is voluntary?

☐ Yes

☐ No

☒ N/A

Will you obtain written consent for participation?

☐ Yes

☐ No

☒ N/A

If the research is observational, will you ask participants for their consent to being observed?

☐ Yes

☐ No

☒ N/A

Will you tell participants that they may withdraw from the research at any time and for any reason?

☐ Yes

☐ No

☒ N/A

With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?

☐ Yes

☐ No

☒ N/A

Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?

☐ Yes

☐ No

☒ N/A

Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?

☐ Yes

☐ No

☒ N/A

Notes: If you have ticked No to any of questions above, you should complete the Full Ethics Approval Form.

Full Ethics Form

Figure 25: Ethics Form 1

Ethics Form (Fast-Track)

1 Participant Confidentiality 2 **Nature of Research** 3 Target Participants 4 Support Information

Nature of Research

Will your project/assignment deliberately mislead participants in any way?	☐ Yes	☐ No	☒ N/A
Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort? This should include details of what you will tell participants to do if they should experience any problems (e.g., who they can contact for help). You may also need to consider risk assessment issues.	☐ Yes	☐ No	☒ N/A
Is the nature of the research such that contentious or sensitive issues might be involved? This includes research which could induce psychological stress, anxiety or humiliation, or cause more than minimal pain.	☐ Yes	☐ No	☒ N/A
Does your research involve the use of sensitive materials? E.g., records of personal or sensitive confidential information.	☐ Yes	☐ No	☒ N/A
Does your research require external agency approval?	☐ Yes	☒ No	☐ N/A
Does your research use hazardous or controlled substance?	☐ Yes	☒ No	☐ N/A
Does your research require you to visit participants in their home or non-public space?	☐ Yes	☐ No	☒ N/A
Does your research investigate illegal activities or behaviours?	☐ Yes	☒ No	☐ N/A
Does your research involve discussion or collection of information on potentially sensitive, embarrassing or distressing topics, administrative or secure data? This includes research involving respondents through internet where visual images are used, and where sensitive issues are discussed	☐ Yes	☒ No	☐ N/A
Will your participants be receiving financial compensation for participating in your research?	☐ Yes	☐ No	☒ N/A
Will your research data be used in the future after the conclusion of your project?	☐ Yes	☒ No	☐ N/A
Will your research involve in processing sensitive data belonging to an organisation/persons?	☐ Yes	☒ No	☐ N/A
Will your research be collecting photographs, videos, and audio recordings of the participants?	☐ Yes	☐ No	☒ N/A
Will the participants' personal particulars be known to any third party?	☐ Yes	☐ No	☒ N/A
Will the participants' data confidentiality be made known to the public?	☐ Yes	☐ No	☒ N/A
Will the research be conducted where the safety of the researchers maybe in question?	☐ Yes	☒ No	☐ N/A
Will be the research be conducted outside of Malaysia and/or UK?	☐ Yes	☒ No	☐ N/A

Note: If you have ticked YES to any of questions above, you should complete the Full Ethics Approval Form.
Full Ethics Form

Figure 26: Ethics Form 2

Supervisor Remarks

Name	Remarks	Date
ASSOC. PROF. DR. NIRASE FATHIMA ABUBACKER	Acceptable	Apr 17, 2026, 4:34:52 AM

Ethics Form (Fast-Track)

1

Participant Confidentiality

2

Nature of Research

3

Target Participants

4

Support Information

Target Participants

Do participants fall into any of the following special groups?

- Children (under 18 years of age)
- People with communication or learning difficulties
- Patients
- People in custody
- People who could be regarded as vulnerable
- People engaged in illegal activities (e.g., drug taking)
- Groups of people whose relationship among each other allow one to have influence over the other such as: Carers and patients with chronic conditions; teachers and their students; prison authorities and prisoners; employers and employees
- Groups where permission of a gatekeeper is normally required for initial access to members.

Note: You may also need to obtain satisfactory clearance from the relevant authorities.

Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?

Note: If you have ticked Yes to any of questions above, you should complete the Full Ethics Approval Form. There is an obligation on student and supervisor to bring to the attention of the APU School Research Ethics Committee any issues with ethical implications not clearly covered by the above checklist.
Full Ethics Form

Prev

Next

Figure 27: Ethics Form 3

Supervisor Remarks		
Name	Remarks	Date
ASSDC, PROF. DR. NIRASE FATHIMA ABUBACKER	Acceptable	Apr 17, 2026, 4:34:52 AM

Ethics Form (Fast-Track)			
1	2	3	4
Participant Confidentiality	Nature of Research	Target Participants	Support Information

Support Information	
I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee.	☐
I am aware of APU liability policy and will make the necessary arrangement for insurance coverage of all researchers and participants of the project/assignment.	☒
Description	This project uses nine publicly available open source government datasets. These include NYC Taxi and Limousine Commission yellow and green taxi trip records, MTA subway station data, MTA commuter rail station data, US Census ACS data covering population density, median household income, and car ownership at the census tract level, and NOAA Central Park hourly weather data. All datasets are anonymised and aggregated with no personally identifiable information. The data is freely accessible from official government portals, including NYC Open Data, the US Census Bureau data portal, and the NOAA National Centers for Environmental Information. No human participants, surveys, interviews, or primary data collection are involved. The research involves
Consent Form (Maximum 3 files)	
Information Sheet	
Additional Files	
I also confirm that:	
Note: Please check either 1 of the checkboxes.	
i) All key documents e.g., consent form, information sheet, questionnaire/interview, and all material such as emails and posters for the purpose of recruitment of participants are appended to this application.	☐
ii) Any key documents e.g. consent form, information sheet, questionnaire/interview schedules which need to be finalised following initial investigations will be submitted for approval by the project/assignment supervisor/module lecturer before they are used in primary data collection.	☒

Figure 28: Ethics Form 4

Appendix C: Log Sheets (3 log sheets)

Meeting Log	
Title	Optimizing First-Mile Last-Mile Transit Routes Using Machine Learning for Demand Prediction in Klang Valley
Supervisor Name	nfathima.abubacker
Created At	21-04-2026 03:17 PM
Meeting Name	Meeting 1
Meeting Schedule Date	30-03-2026 12:00 pm
Location	ONLINE MEETING
Content	
Items For Discussion	Review complete Chapter 1 for alignment between problem statements, aim, objectives, and scope Review Chapter 2 LR summary (Sections 2.5.1 to 2.5.3) to confirm all gaps are clearly articulated Review Chapter 3 methodology for completeness: CRISP-DM phase explanations, dataset descriptions, preprocessing pipeline, and initial data understanding Review Chapter 4 for achievement discussion, SDG contribution, and research gaps for Semester 2 Confirm Gantt chart follows CRISP-DM phases as the main tasks Review appendix sections: PPF screenshot, ethics form screenshot, and log sheets Check Turnitin similarity and AI detection percentage Final formatting check: cover page, table of contents, page numbers, APA references, abstract word count
Content of Discussion	Chapter 1 was confirmed as complete; supervisor noted that the scope section is one of the strongest parts of the IR because it clearly separates what is included, what is excluded, and why Chapter 2 LR summary was approved; supervisor confirmed that Section 2.3.2 clearly identifies all three gaps and directly maps them to the three problem statements in Chapter 1 Chapter 3 CRISP-DM implementation explanation in Section 3.2.3 was reviewed; supervisor advised adding the expected output for each phase so the second marker can map it to the Gantt chart Chapter 4 was reviewed; supervisor confirmed the achievement discussion adequately connects the investigation findings to SDG 11.2 and clearly identifies what remains for Semester 2 Gantt chart was confirmed as following CRISP-DM phases (Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation); supervisor approved it as correctly structured Turnitin similarity was below 20% and AI detection was below 30%; both are within acceptable range Abstract word count confirmed at 247 words, within the 200 to 250 word guideline; one sentence was trimmed to bring it within range APA references were checked; supervisor flagged two references missing the DOI and advised a
Action List	Add expected output column or sentence to each CRISP-DM phase description in Section 3.2.3 Add missing DOIs to the two flagged references in the reference list Screenshot the submitted PPF from FYP Bank and insert into Appendix A Screenshot the approved ethics form from FYP Bank and insert into Appendix B Fill in all three project log sheets on FYP Bank and screenshot them for Appendix C Submit final IR on Moodle in Word document format before 11:59 PM on 22 April 2026 Remind supervisor to approve all three meeting requests on FYP Bank so log sheets can be accessed

Figure 29: Meeting Logs 1

Meeting Log	
Title	Optimizing First-Mile Last-Mile Transit Routes Using Machine Learning for Demand Prediction in Klang Valley
Supervisor Name	nfathima.abubacker
Created At	21-04-2026 03:15 PM
Meeting Name	Meeting 2
Meeting Schedule Date	10-04-2026 11:00 am
Location	ONLINE MEETING

Content	
Items For Discussion	Review domain research structure: four research domains covering urban transport, machine learning approaches, feature engineering, and model evaluation Discuss similar works comparison table (seven studies including Correa & Moyano 2023, Zhao et al. 2021, Liu & Chen 2026) Confirm that CRISP-DM is the appropriate methodology for this CSDA project Discuss the five datasets selected and whether they are open-source and legally usable Review the data preprocessing pipeline: null handling, invalid trip removal, temporal feature extraction, spatial joins, and zone-hour aggregation Confirm the enriched feature set: subway station proximity and car ownership rate from Census ACS Table 808141 Clarify the chronological 80/20 train-test split approach
Content of Discussion	Supervisor approved the four-domain structure for Chapter 2 and confirmed that the SDG must be included as one of the five minimum research domains, not as a standalone section The similar works comparison table was reviewed: supervisor noted that Liu and Chen (2026) must be clearly described as a resilience framework with no prediction model, so that the gap this project fills is obvious to the second marker CRISP-DM was confirmed as the correct methodology; supervisor advised including a comparison table between CRISP-DM, KDD, and SEMMA with a justification paragraph below the table All five datasets were confirmed as legally open-source; supervisor advised explicitly stating the data licence or access condition for the Census ACS data in Chapter 3 Supervisor flagged that the straight-line distance measure for subway proximity must acknowledge its limitation (it ignores road networks and physical barriers) and that this should appear in both the scope constraints and the methodology The 80/20 chronological split was confirmed and must be justified by referencing Correa and Moyano (2023) for direct comparability Supervisor confirmed that default hyperparameters must be applied to all four model runs and that no tuning should occur
Action List	Add CRISP-DM versus KDD versus SEMMA comparison table to Chapter 3 Section 3.2 with a justification paragraph selecting CRISP-DM Add the data licence or access condition for Census ACS Table 808141 in Chapter 3 Section 3.3 Update the similar works summary in Section 2.3.2 to explicitly state that no existing study combines zone-level accuracy with transit accessibility in a single tool Ensure the straight-line distance limitation for subway proximity appears in Section 1.5.2.3 and is cross-referenced in Chapter 3 Complete all five research domain subsections in Chapter 2 with a minimum of three citations per subtopic Draft the data collection subsection (Section 3.3) with source links and justification for each of the five datasets

Figure 30: Meeting Log 2

Meeting Log	
Title	Evaluating Machine Learning Prediction Performance Across New York City Taxi Zones to Identify Underserved Communities
Supervisor Name	nfathima.abubacker
Created At	21-04-2026 03:13 PM
Meeting Name	Meeting 4
Meeting Schedule Date	20-04-2026 12:00 am
Location	ONLINE MEETING

Content	
Items For Discussion	Confirm FYP title: Evaluating Machine Learning Model Performance for Taxi Demand Prediction in Underserved Urban Zones of New York City Using Zone-Stratified Analysis and Multi-Source External Features Review and validate the three problem statements (lack of zone-level evaluation, absence of contextual features, lack of actionable planning tool) Confirm project aim, four objectives, and scope boundaries Confirm SDG 11 (Target 11.2) mapping and justification Clarify what is excluded from scope (green taxis, LSTM, ride-hailing data, zone ID as a feature) Confirm target users (transport planners and data science researchers)
Content of Discussion	Supervisor confirmed the FYP title is clear and follows the required structure (verb, focus, domain, population) Problem Statement 1 was validated: the critique of Correa and Moyano (2023) for reporting only aggregate R^2 without zone-level breakdown is academically sound and well-supported Problem Statement 2 was validated: the absence of subway proximity and car ownership features in existing models is a genuine gap acknowledged even by the anchor paper itself Problem Statement 3 was validated: the absence of an actionable dashboard combining model performance with transit accessibility is a meaningful contribution Supervisor confirmed that excluding Zone ID as a model feature is the correct methodological decision and must be clearly justified in both Chapter 1 and Chapter 2 SDG 11 Target 11.2 mapping was approved; supervisor advised adding one sentence of explanation for why Target 11.3 is also referenced in the scope Scope boundaries were agreed upon; supervisor noted that the constraint on default hyperparameters must be explicitly justified in Chapter 2 before Chapter 3 references it
Action List	Revise the scope section to add a brief justification for why Target 11.3 is mentioned alongside Target 11.2 Add a forward reference in Section 1.5.2.2 pointing to Chapter 2 Section 2.2.4.3 for the hyperparameter justification Begin drafting Chapter 2 domain research, starting with the urban transport and taxi demand section Ensure every problem statement in Chapter 1 ends with a citation-supported closing sentence

Figure 31: Meeting Logs 3

Appendix D: Gantt Chart

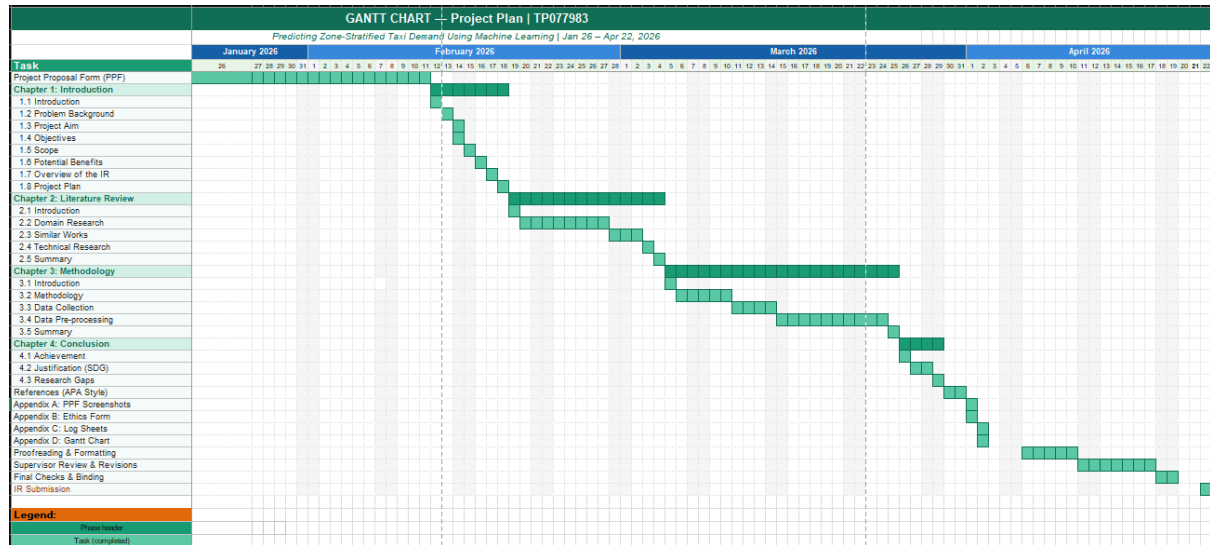


Figure 32: Gantt Chart

